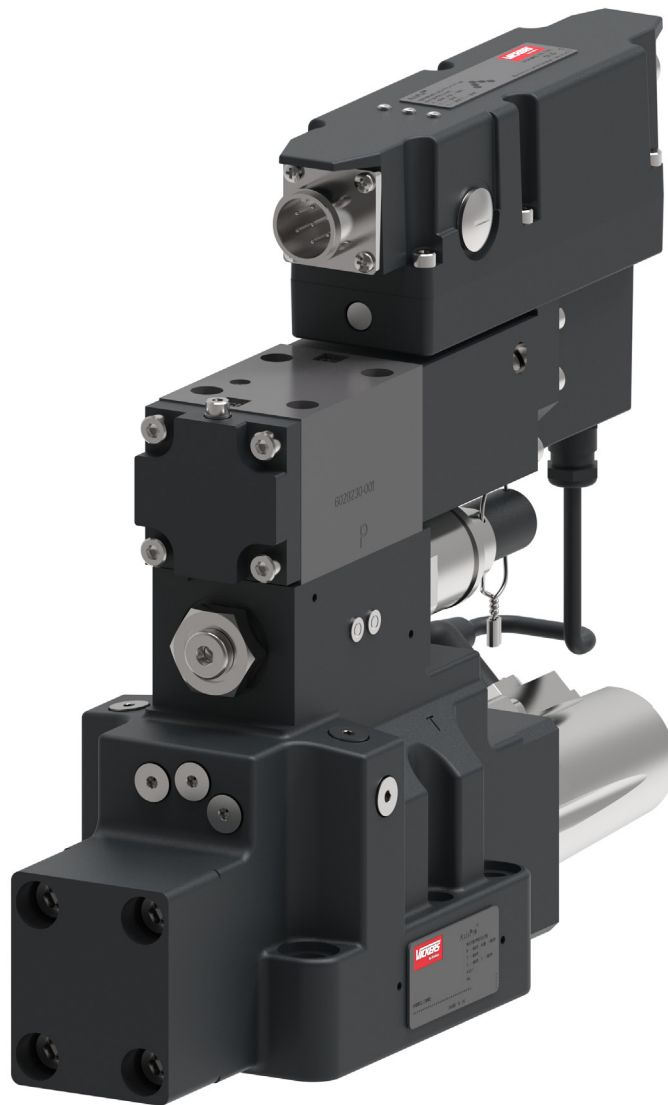


# AxisPro® KBHD15-07, -01 Series

## Servo-Performance Proportional Directional Valve



## Contents

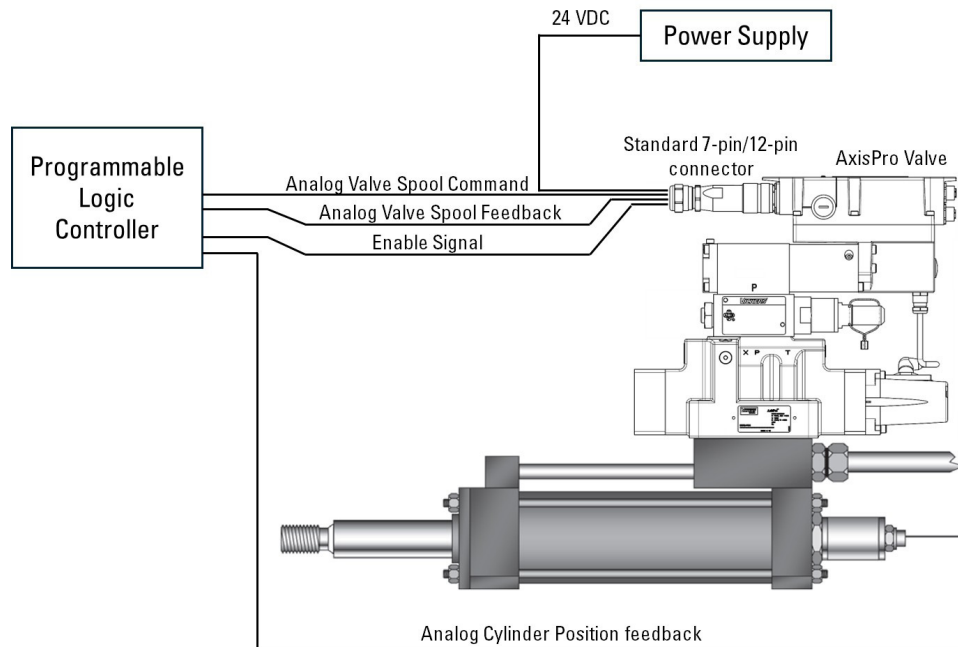
|                                                                                               |           |
|-----------------------------------------------------------------------------------------------|-----------|
| <b>1. Introduction .....</b>                                                                  | <b>3</b>  |
| 1.1. General Description .....                                                                | 3         |
| 1.2. Typical Section View .....                                                               | 4         |
| <b>2. Model Code .....</b>                                                                    | <b>5</b>  |
| <b>3. Spool Details .....</b>                                                                 | <b>6</b>  |
| <b>4. Performance Curves .....</b>                                                            | <b>9</b>  |
| <b>5. Operating Data .....</b>                                                                | <b>11</b> |
| 5.1. Connector Details .....                                                                  | 12        |
| 5.2. LED Diagnostic Details (Location 5) .....                                                | 13        |
| 5.3. DIP Switch: Baud rate & node ID setting (Location 2).....                                | 16        |
| 5.4. MicroUSB serial communication (Location 2) .....                                         | 16        |
| 5.5. Electromagnetic compatibility (EMC) .....                                                | 17        |
| 5.6 Operating Conditions.....                                                                 | 17        |
| <b>6. Software Information .....</b>                                                          | <b>19</b> |
| <b>7. Electrical Information .....</b>                                                        | <b>20</b> |
| 7.1. Block Diagram Voltage Input (Field 5=1) .....                                            | 20        |
| 7.2. Block Diagram Current Input (Field 5=2,3,4,5) .....                                      | 21        |
| 7.3. Wiring Connections Voltage Input (Field 5 = 1) .....                                     | 22        |
| 7.4. Wiring Connections for Voltage Mode (Field 5 = 1) Valves with Enable Feature .....       | 23        |
| 7.5. Wiring Connections Current Input (Field 5 = 2,3,4,5).....                                | 24        |
| 7.6. Wiring Connections for Current Input (Field 5 = 2,3,4,5) Valves with Enable Feature..... | 25        |
| 7.7. Wiring Connection of Valve in Current Mode Command with current as input.....            | 26        |
| 7.8. Wiring Connection of Valve in Current Mode Command with voltage as input.....            | 27        |
| 7.9. Warning .....                                                                            | 27        |
| <b>8. Installation Dimensions. ....</b>                                                       | <b>28</b> |
| <b>9. Mounting Surfaces .....</b>                                                             | <b>30</b> |
| <b>10. Application Data .....</b>                                                             | <b>31</b> |



This product has been designed and tested to meet specific standards outlined in the European Electromagnetic Compatibility Directive (EMC) 2014/30/EU which repealed Directive 2004/108/EC. For Restriction of Hazardous Substances, complies to (RoHS) Directive 2011/65/EU including amendment 2015/863.

## 1. Introduction

AxisPro is a game changing machine control valve. Its embedded intelligence simplifies traditionally complex control practices. Plug and play design reduces machine build time, and its ability to predict potential maintenance issues increases machine reliability.



AxisPro level 1 valves, such as KBHD15-07 (ISO Size 07), can be used to control machine motions in open loop or closed loop control applications. The valve receives its analog command input on the 7-pin or 12-pin main connector from an external axis control device.

### 1.1 General Description

Built on the proven KBH servo Proportional Valve technology, the AxisPro Proportional Valve provides a range of control capabilities in a modular design. These solenoid operated proportional valves offer high dynamic performance which enables them to be used in closed-loop control applications.

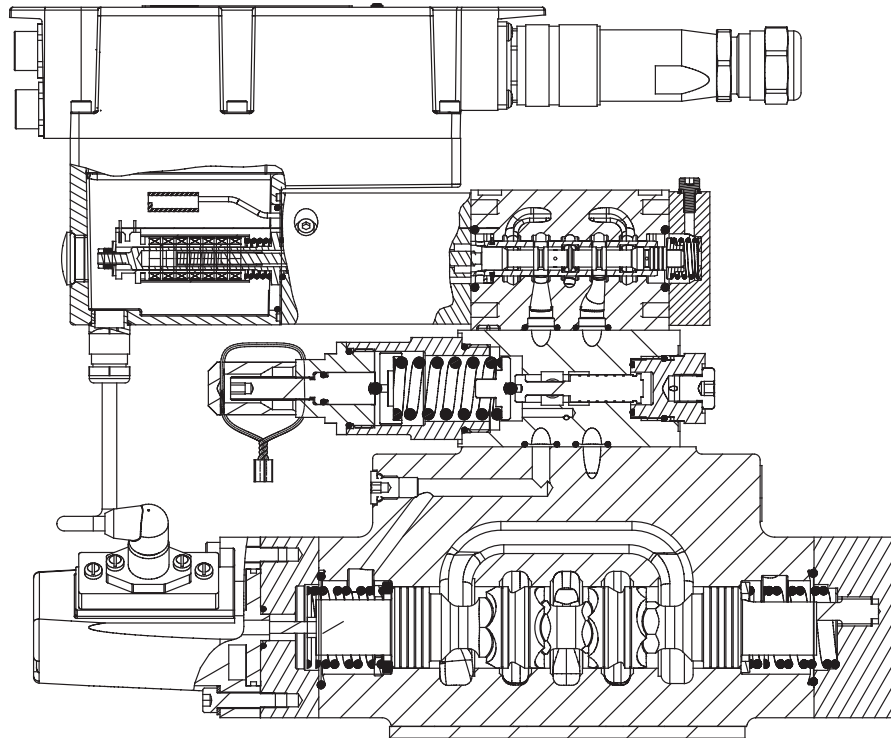
Best-in-class ingress protection, rated to IP65 and IP67, combined with upto 85°C (185°F) ambient temperature, allows operation in demanding environments.

## 1.2 Unique benefits from AxisPro

Reliable, extended uptime is enabled by valve and systems diagnostics capability. LEDs provides on-valve diagnostics information for valves. Access to systems and machine health data can be made available via CANopen and Ethernet networked valves.

Leverage inventory of AxisPro valves by configuration through software. One valve SKU can serve multiple needs: Valves can be configured via Danfoss's Pro-FX™ Configure software tool for optional command signal: Voltage or current, as well as activating the "enable"-pin. Valves can also have CAN and Industrial Ethernet bus activated and control modes selected and configured.

*Typical Section View of KBHD15-07*



## 2. Model Code

| KBHD15 | 07 | ***** | N | * | * | *** | ** | N | 01 | H  | 001 | 00 | ***** |
|--------|----|-------|---|---|---|-----|----|---|----|----|-----|----|-------|
| 1      | 2  | 3     | 4 | 5 | 6 | 7   | 8  | 9 | 10 | 11 | 12  | 13 | 14    |

### 1 Valve Type

**KBHD15** – Two stage servo performance proportional valve with integral amplifier and electronic feedback

### 2 Interface

**07** – ISO 4401, Size 07

### 3 Spool/Sleeve

**000002C230N** – 230 L/min symmetric, overlapped, P, A, B, T blocked at null

**000033C230N** – 230 L/min symmetric, P blocked, A & B connected to tank at null

**000002C200N** – 200 L/min symmetric, overlapped, P, A, B, T blocked at null

**000033C160N** – 160 L/min symmetric, P blocked, A & B connected to tank at null

**000005C230N** – 230 L/min symmetric, zero lapped

**002C230N140** – 230 L/min / 140 L/min, overlapped, P, A, B, T blocked at null

**033C230N140** – 230 L/min / 140 L/min, overlapped, P blocked, A & B connected to tank at null

**0002C150N85** – 150 L/min / 85 L/min, overlapped, P, A, B, T blocked at null

**0033C130N65** – 130 L/min / 65 L/min, overlapped, P blocked, A & B connected to tank at null

**005C230N140** – 230 L/min / 140 L/min, zero lapped

**133C230N140** – 230 L/min / 140 L/min, overlapped, P blocked, A & B connected to tank at null

**535C190N110** –

**00000PQ230F** – Pressure flow control spool

**000PQ230F50** – Pressure flow control spool

**003PQ230F50** – Pressure flow control spool

### 4 Pilot supply & drain option

**A** – Internal pilot with reducer, internal drain

**B** – Internal Pilot without reducer, Internal drain

**C** – Internal pilot without reducer, external drain

**D** – External pilot without reducer, internal drain

**E** – External pilot without reducer, external drain

**F** – External pilot with reducer, external drain

**G** – Internal pilot with reducer, external drain

**H** – External pilot with reducer, internal drain

### 5 Command Signal

**1** +/- 10V voltage command

**2** 4-20mA current command

**3** +/-10mA current command

**4** +/- 15mA current command

**5** +/- 20mA current command

### 6 Monitor Output

**1** +/- 10V voltage feedback signal

**2** 4-20mA current feedback signal

### 7 Electrical connection

**C07** – 7 pin connector without plug, no enable

**E07** – 7 pin connector with plug, no enable

**H07** – 7 pin connector with plug, pin “C” enable

**R07** – 7 pin connector without plug, pin “C” enable

**C12** – 12 pin connector without plug, no enable

**E12** – 12 pin connector with plug, no enable

**H12** – 12 pin connector with plug, pin “C” enable

**R12** – 12 pin connector without plug, pin “C” enable

### 8 Digital communication interface

**NS** – No option

**CO** – CANopen

**PN** – Profinet

**EI** – Ethernet/IP

**EC** – EtherCAT

### 9 External sensor

**N** – No option

### 10 Design Series

**01** – 01 series, subject to change

### 11 Coil Rating

**H** – 24V DC Amplifier supply

### 12 Software revision

**001** – Software revision

### 13 Seal option

**00** – FKM Seals

**F1** – NBR Seals

### 14 Special Feature

**00000** – Not applicable

**CR720** – 720 hours of corrosion protection

**VR000** – Standard corrosion protection, high vibration resistance

**VR720** – 720 hours of corrosion protection, high vibration resistance

**ENXXX** – For any special feature requirements, contact Danfoss for new EN special.

## Customer Configuration

The following features can be configured by the customer:

[4] Internal / External pilot supply: Install or remove set screw (Page 29)

[4] Internal / External pilot drain: Install or remove set screw (Page 29)

[5] Command signal: All options 1 to 5 can be changed via ProFx

[6] Monitor output: All options 1 to 2 can be changed via ProFx

[7] C/E E/H: Enable options can be changed via ProFx

[7] With and without connector: The connector can be purchased separately

[8] Digital communication interface: Ethernet protocols PN/EI/EC can be exchanged via ProFx

### 3. Spool Details

| Spool/Sleeve | Hydraulic symbol | Description                      | Flow<br>ℓ/min@ ΔP<br>10 bar | Symmetric | Asymmetric |
|--------------|------------------|----------------------------------|-----------------------------|-----------|------------|
| 2C230N       |                  | Overlapped,<br>All Ports blocked | 230                         | ✓         | -          |
| 33C230N      |                  | Overlapped,<br>A,B,T connected   | 230                         | ✓         | -          |
| 2C200N       |                  | Overlapped,<br>All Ports blocked | 200                         | ✓         | -          |
| 33C160N      |                  | Overlapped,<br>A,B,T connected   | 160                         | ✓         | -          |
| 5C230N       |                  | Critically lapped                | 230                         | ✓         | -          |
| 2C230N140    |                  | Overlapped,<br>All Ports blocked | 230/140                     | -         | ✓          |
| 33C230N140   |                  | Overlapped,<br>A,B,T connected   | 230/140                     | -         | ✓          |
| 2C150N85     |                  | Overlapped,<br>All Ports blocked | 150/85                      | -         | ✓          |
| 33C130N65    |                  | Overlapped,<br>A,B,T connected   | 130/65                      | -         | ✓          |
| 5C230N140    |                  | Critically lapped                | 230/140                     | -         | ✓          |
| 133C230N140  |                  | Overlapped,<br>A,B,T connected   | 230/140                     | -         | ✓          |
| 535C190N110  |                  |                                  |                             |           | ✓          |

| Spool/Sleeve | Hydraulic symbol | Description     | Flow<br>ℓ/min@ ΔP<br>10 bar | Symmetric | Asymmetric |
|--------------|------------------|-----------------|-----------------------------|-----------|------------|
| PQ230F       |                  | See flow curves | 230                         | ✓         | -          |
| PQ230F50     |                  | See flow curves | 230/50                      | -         | ✓          |
| 3PQ230F50    |                  | See flow curves | 230/50                      | -         | ✓          |

### Application Notes

#### 1. Main-Spool Options

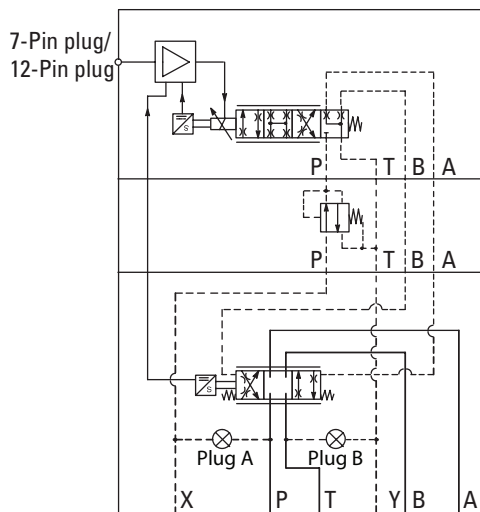
Spools shown are meter-in/meter-out types. Center-condition options are type 2, 33, 535 and PQ.

#### 2. Internally Piloted Models

Differ from detailed symbols below by omission of plug A and the blocking of port X by the mating surface.

#### 3. Internally Pilot Drain Models

Differ from detailed symbols below by omission of plug B and blocking of port Y by the mating surface.



## Spool Types and Flow Ratings

### Symmetric Spools

Base line pressure drop  $\Delta p$  5 bar (72 psi) per metering flow path, e.g. B to T. For actual maximum flow refer to power capacity envelope curves.

**Note:** Valves with critically lapped spool are designed so that with the valve disabled the pressure in port B is at least twice that in port A (blocked ports).

### Asymmetric spools

Figure preceding metering type designator, "N" e.g. 2C\*\*N is flow rating P-A or A-T ("A" port flow).

Figure after "N" (N\*\*) is flow rating P-B or B-T ("B" port flow)

## Pressure and Minimum Flow Rates

*Maximum Pressures, Bar (PSI) Valves With Pressure Reducer*

| Model     | Pilot pressure source † | Pilot drain connection | P Port    | A&B Ports | T Port    | X Port ♦  | Y Port  |
|-----------|-------------------------|------------------------|-----------|-----------|-----------|-----------|---------|
| KBHD15-07 | External                | External               | 350(5000) | 50(5000)  | 350(5000) | 350(5000) | 50(700) |
|           |                         | Internal*              | 350(5000) | 350(5000) | 50(700)   | 350(5000) | 50(700) |
|           | Internal                | External               | 350(5000) | 350(5000) | 350(5000) | 350(5000) | 50(700) |
|           |                         | Internal*              | 350(5000) | 350(5000) | 50(700)   | 350(5000) | 50(700) |

† - Minimum recommended pilot operating pressure = 50 bar (700 psi)

\* - Internal drain is a non-preferred option

♦ - For pilot pressures  $\leq$  210 bar (3000 psi) a pilot pressure reducer is optional.  
For pilot pressures  $>$  210 bar (3000 psi) a pilot pressure reducer must be used

 - Unused pilot port: Maximum pressure as shown

*Maximum Pressures, Bar (PSI) Valves Without Pressure Reducer*

| Model     | Pilot pressure source † | Pilot drain connection | P Port    | A&B Ports | T Port    | X Port ♦  | Y Port  |
|-----------|-------------------------|------------------------|-----------|-----------|-----------|-----------|---------|
| KBHD15-07 | External                | External               | 350(5000) | 350(5000) | 350(5000) | 210(3000) | 50(700) |
|           |                         | Internal*              | 350(5000) | 350(5000) | 50(700)   | 210(3000) | 50(700) |
|           | Internal                | External               | 210(3000) | 350(5000) | 350(5000) | 210(3000) | 50(700) |
|           |                         | Internal*              | 210(3000) | 350(5000) | 50(700)   | 210(3000) | 50(700) |

† - Minimum recommended pilot operating pressure = 50 bar (700 psi)

\* - Internal drain is a non-preferred option

♦ - For pilot pressures  $\leq$  210 bar (3000 psi) a pilot pressure reducer is optional.  
For pilot pressures  $>$  210 bar (3000 psi) a pilot pressure reducer must be used

 - Unused pilot port: Maximum pressure as shown

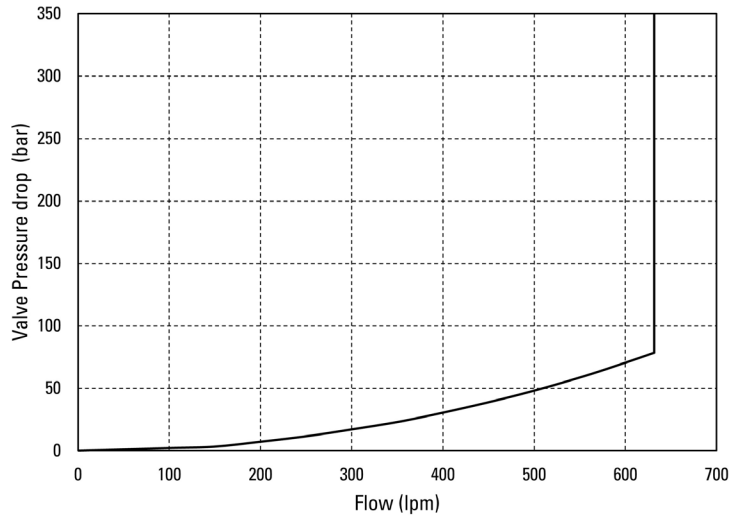
*Minimum Recommended Flow Rates*

| Valve size/Spool code | Min. flow rate [L/min] | in <sup>3</sup> /min |
|-----------------------|------------------------|----------------------|
| KBHD15-07             | 1.0                    | 60                   |

## 4. Performance Curves

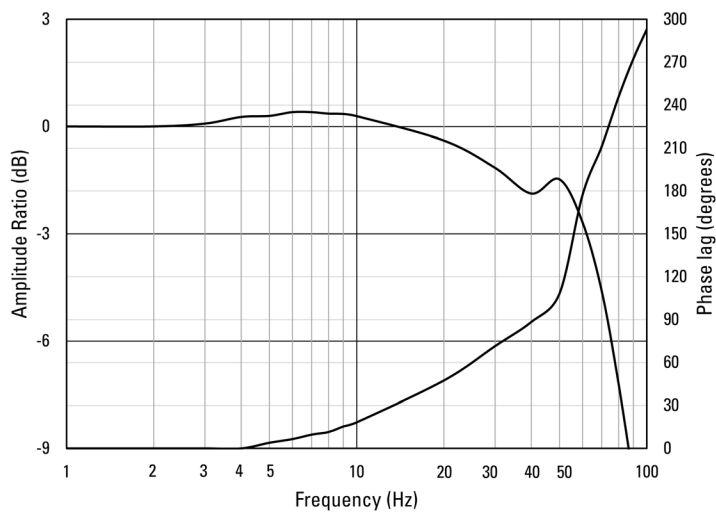
### Power capacity envelope

Flow through P-A-B-T or P-B-A-T



### Frequency response, Typical

For an amplitude of  $50 \pm 25\%$  of rated flow (ISO 10770-1) 2C230N spool measured at  $v = 36$  cSt (168 SUS),  $t = 50^\circ\text{C}$  (122°F) and pilot pressure = 40 bar (with-reducer model)

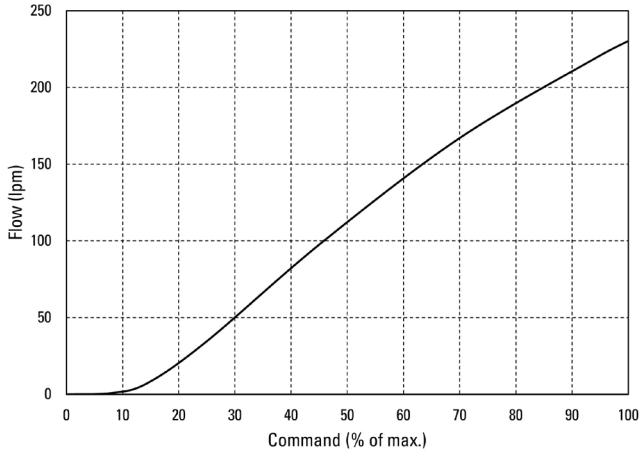


## Flowgain

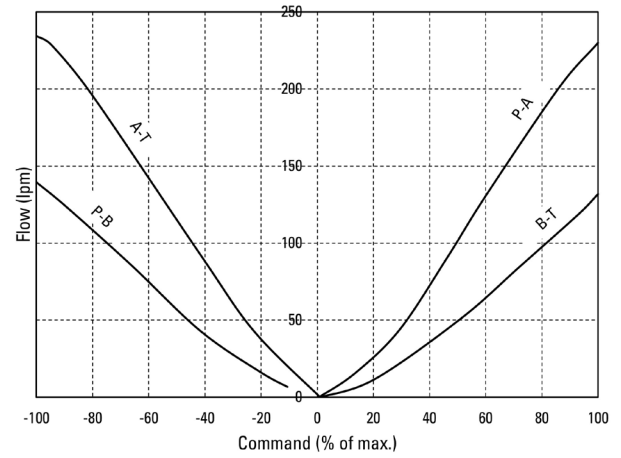
At  $\Delta P = 5$  bar (72psi) per metering flow path (e.g. P-A)

Percentage command signals applicable for positive and negative values of command signal.

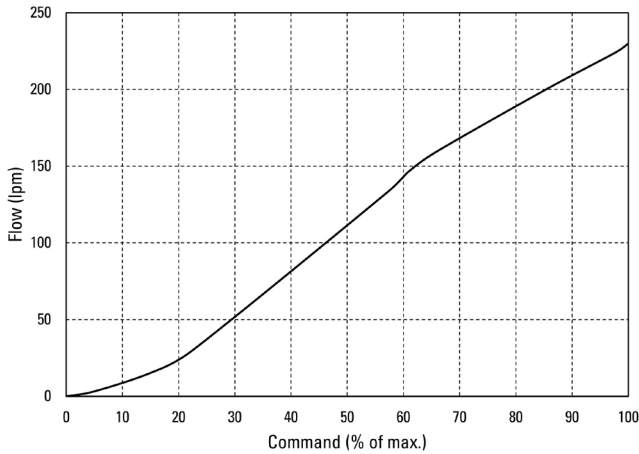
2C230N



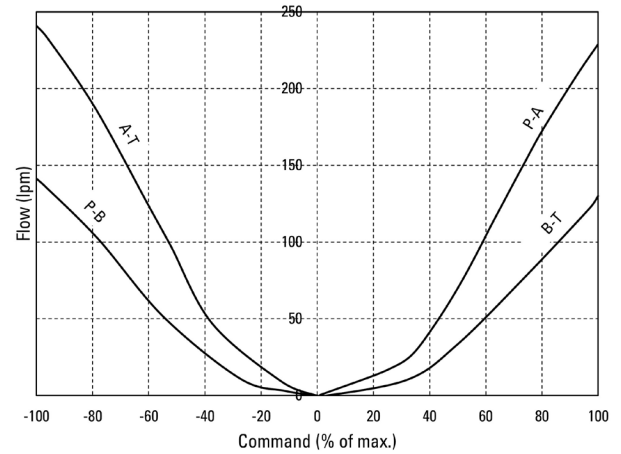
2C230N140



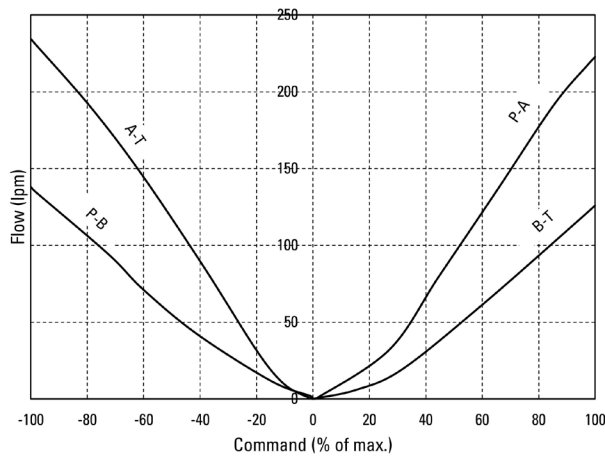
5C230N



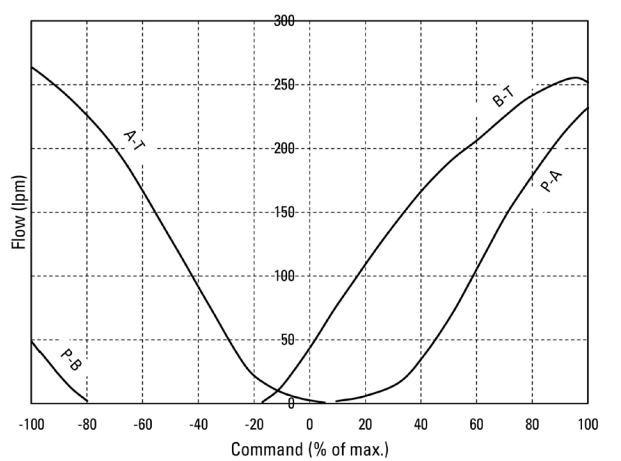
5C230N140



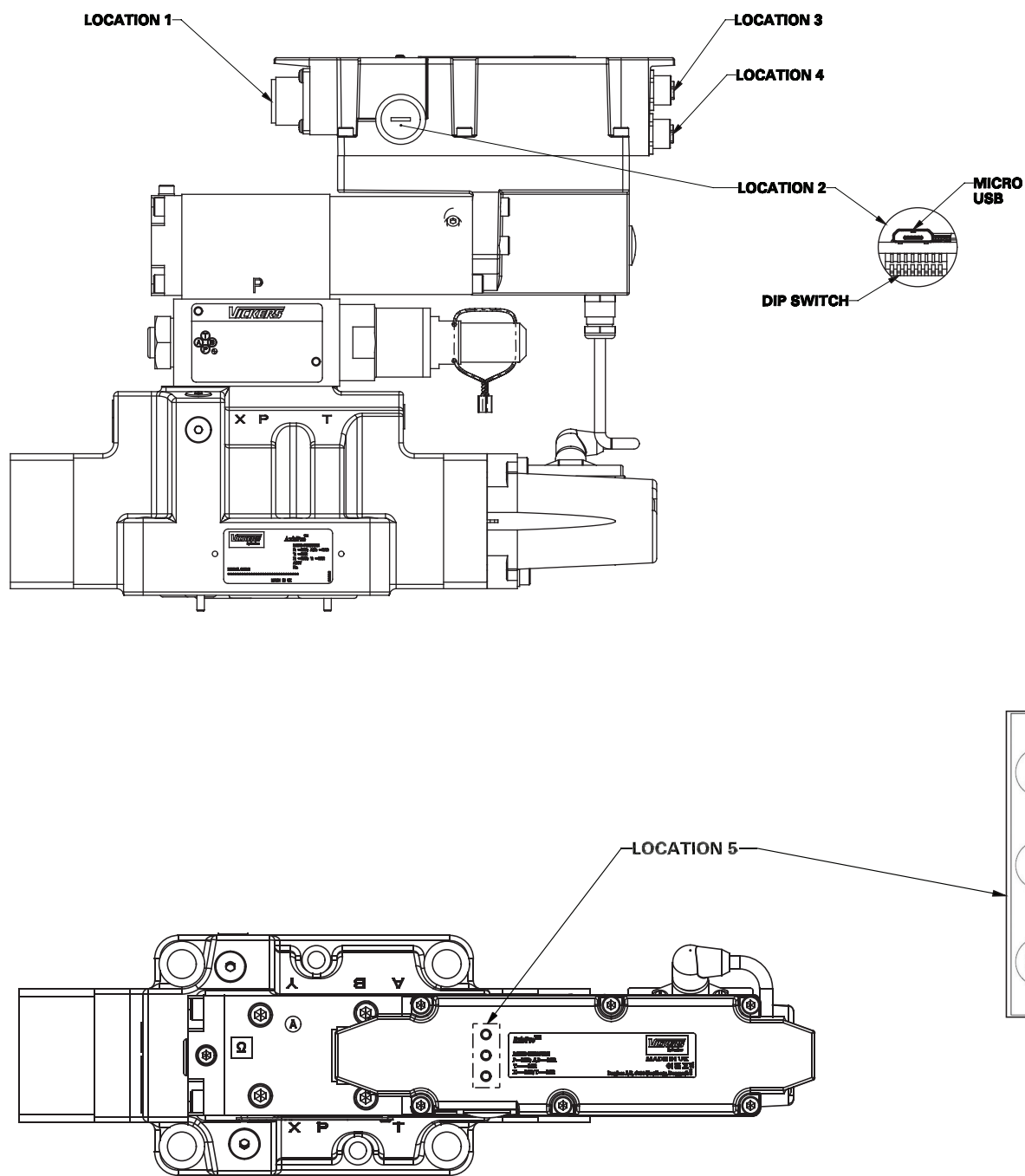
33C230N140



PQ230F50

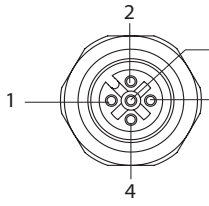
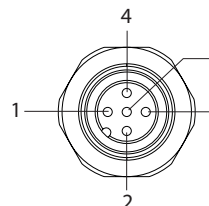
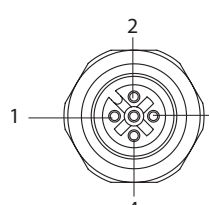


## 5. Operating Data



### 5.1 Connector Details

| Connector                           | Image | Pin | Signal                             | Technical specification                                                                                           | Note                                                                                                                           |
|-------------------------------------|-------|-----|------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| LOCATION 1<br>7-pin plug connector  |       | A   | Power supply positive (+)          | Supply voltage: 18-36 V DC                                                                                        | To ensure EMI protection use only metal shielded mating connectors. Mating 7-pin (connector) is Danfoss part number 934939     |
|                                     |       | B   | Power supply 0V ground return      | Power supply ground (0V)                                                                                          |                                                                                                                                |
|                                     |       | C   | Not connected / Valve enable       | Enable Signal >12V (max 36V)                                                                                      |                                                                                                                                |
|                                     |       | D   | Command signal (+V or current in)  | Voltage mode: $\pm 10V$ DC<br>Current mode: 4-20 mA or $\pm 20mA$                                                 |                                                                                                                                |
|                                     |       | E   | Command signal return              | Command signal (-V or current GND)                                                                                |                                                                                                                                |
|                                     |       | F   | Monitor output                     | Voltage mode: $\pm 10V$ DC,<br>Current mode: 4-20 mA                                                              |                                                                                                                                |
|                                     |       | G   | Protective earth                   | Internally connected to housing                                                                                   |                                                                                                                                |
| LOCATION 1<br>12-pin plug connector |       | 1   | +24V Drive stage supply            | Supply Voltage: 18-36 V DC                                                                                        | To ensure EMI protection use only metal shielded mating connectors. Mating 12-pin (connector) is Danfoss part number 12007132. |
|                                     |       | 2   | Power supply 0V                    | Power supply ground                                                                                               |                                                                                                                                |
|                                     |       | 3   | Valve enable                       | Enable Signal >12V (max 36V)                                                                                      |                                                                                                                                |
|                                     |       | 4   | Command signal (+V or current in)  | Voltage mode: $\pm 10V$ DC,<br>Current mode: 4-20 mA or $\pm 20mA$ DC                                             |                                                                                                                                |
|                                     |       | 5   | Command signal (-V or current GND) | Command signal (-V or current GND)                                                                                |                                                                                                                                |
|                                     |       | 6   | Monitor output                     | Voltage mode: $\pm 10V$ DC,<br>Current mode: 4-20 mA                                                              |                                                                                                                                |
|                                     |       | 7   | NC                                 |                                                                                                                   |                                                                                                                                |
|                                     |       | 8   | Enable acknowledgement (24V)       | Fault-free operation: 0V<br>Error: 24V<br>(Equivalent to input power supply)<br>max current: 50mA                 |                                                                                                                                |
|                                     |       | 9   | +24V Power supply Electronics      | Supply Voltage: 18-36 V DC<br>Max. ripple: 10% peak-to-peak<br>(comply with absolute supply voltage limit values) |                                                                                                                                |
|                                     |       | 10  | GND (0 V)                          | Power supply ground (0V)                                                                                          |                                                                                                                                |
|                                     |       | 11  | Error output signal (24 V)         | Fault-free operation: 0V<br>Error: 24V<br>(Equivalent to input power supply)<br>max current: 1.5A                 |                                                                                                                                |
|                                     |       | 12  | Protective earth                   | Internally connected to housing                                                                                   |                                                                                                                                |

| Connector                                                                                         | Image                                                                             | Pin | Description     | Note                                                                                                                                   |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>LOCATION 3</b><br>M12 5-pin CAN Connector (Female)                                             |  | 1   | CAN shield      | To ensure EMI protection use only metal shielded mating connectors. Use only shielded twisted pair (STP) cables for mating connection. |
|                                                                                                   |                                                                                   | 2   | Not Connected   |                                                                                                                                        |
|                                                                                                   |                                                                                   | 3   | Power supply 0V |                                                                                                                                        |
|                                                                                                   |                                                                                   | 4   | CAN High        |                                                                                                                                        |
|                                                                                                   |                                                                                   | 5   | CAN Low         |                                                                                                                                        |
| <b>LOCATION 4</b><br>M12 5-pin CAN Connector (Male)                                               |  | 1   | CAN shield      | To ensure EMI protection use only metal shielded mating connectors. Use only shielded twisted pair (STP) cables for mating connection. |
|                                                                                                   |                                                                                   | 2   | Not Connected   |                                                                                                                                        |
|                                                                                                   |                                                                                   | 3   | Power supply 0V |                                                                                                                                        |
|                                                                                                   |                                                                                   | 4   | CAN High        |                                                                                                                                        |
|                                                                                                   |                                                                                   | 5   | CAN Low         |                                                                                                                                        |
| <b>LOCATION 3 - PORT 0</b><br><b>LOCATION 4 - PORT 1</b><br>M12 4-pin Ethernet Connector (Female) |  | 1   | TxD+            | M12 connector is D-Coded<br>To ensure EMI protection use only premade CAT5 or CAT 6 cable.                                             |
|                                                                                                   |                                                                                   | 2   | RxD+            |                                                                                                                                        |
|                                                                                                   |                                                                                   | 3   | TxD-            |                                                                                                                                        |
|                                                                                                   |                                                                                   | 4   | RxD-            |                                                                                                                                        |

## 5.2 LED Diagnostic Details (Location 5)

| Diagnostic                                                                          | Color     | CANopen               | Profinet                 | Ethernet/IP    | EtherCAT      |
|-------------------------------------------------------------------------------------|-----------|-----------------------|--------------------------|----------------|---------------|
|  | [A] Green | CAN Connection status | Port 0 connection status | Module status  | Run           |
|                                                                                     | [A] Red   | Not used              | System error             | Not used       | Not used      |
|                                                                                     | [B] Green | Device status         | Device status            | Device status  | Device status |
|                                                                                     | [B] Red   | Fault Manager         | Fault Manager            | Fault Manager  | Fault Manager |
|                                                                                     | [C] Green | Not used              | Port 1 connection status | Not used       | Not used      |
|                                                                                     | [C] Red   | Communication error   | Communication error      | Network status | Error         |

For detailed information on LED Diagnostics, see below table.

### CANopen LED Indication details

| LED         | State                                              | Description                                     |
|-------------|----------------------------------------------------|-------------------------------------------------|
| LED A GREEN | Flashing (2.5Hz ON/OFF rate, 200ms ON, 200ms OFF)  | Device is in CANopen NMT Pre-Operational state. |
|             | ON                                                 | Device is in CANopen NMT operational state.     |
|             | Single flashing (Sequence of 200ms ON, 1000ms OFF) | Device is in NMT stop state.                    |
| LED A RED   | OFF                                                | Not used                                        |

## CANopen LED Indication details (Continued)

| LED         | State                                                                | Description                                                                                  |
|-------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| LED B GREEN | 1Hz frequency (500ms ON and 500ms OFF)                               | No error                                                                                     |
|             | OFF                                                                  | Device error or in boot mode                                                                 |
| LED B RED   | OFF                                                                  | No error                                                                                     |
|             | Blinking (1Hz, 500ms ON 500ms OFF)                                   | Warning faults                                                                               |
|             | Fast blinking (2Hz, 250ms ON and 250ms OFF)                          | Critical faults                                                                              |
| LED C GREEN | OFF                                                                  | Not used                                                                                     |
| LED C RED   | OFF                                                                  | Not used                                                                                     |
|             | Single flashing (Sequence of 200ms ON, 1000ms OFF)                   | At least one of the error counters of the CAN controller has reached the maximum error count |
|             | Double flash (Sequence of 200ms ON, 200ms OFF, 200ms ON, 1000ms OFF) | Node guard or heartbeat error                                                                |

## Profinet LED Indication details

| LED         | State                                       | Description                                                                    |
|-------------|---------------------------------------------|--------------------------------------------------------------------------------|
| LED A GREEN | OFF                                         | The device has no link to the port 0                                           |
|             | Flickering                                  | The device sends/receives frames port 0                                        |
| LED A RED   | OFF                                         | No Error                                                                       |
|             | Flashing (1Hz,3S)                           | DCP signal service is initiated via the bus.                                   |
|             | ON                                          | Watchdog timeout; channel, generic or extended diagnosis present; system error |
| LED B GREEN | 1Hz frequency (500ms ON and 500ms OFF)      | No error                                                                       |
|             | OFF                                         | Device error or in boot mode                                                   |
| LED B RED   | OFF                                         | No error                                                                       |
|             | Blinking (1Hz, 500ms ON 500ms OFF)          | Warning faults                                                                 |
|             | Fast blinking (2Hz, 250ms ON and 250ms OFF) | Critical faults                                                                |
| LED C GREEN | OFF                                         | The device has no link to the port 1                                           |
|             | Flickering                                  | The device sends/receives port 1                                               |
| LED C RED   | OFF                                         | No Error                                                                       |
|             | Flashing(2Hz)                               | No data exchange                                                               |
|             | Single Flash (200ms On and 1000ms Off)      | Device is in Boot mode                                                         |
|             | ON                                          | No configuration; or low speed physical link; or no physical link              |

## Ethernet/IP LED Indication details

| LED         | State                                           | Description                                                                                 |
|-------------|-------------------------------------------------|---------------------------------------------------------------------------------------------|
| LED A GREEN | OFF                                             | LED Module Status No Power                                                                  |
|             | Blink with 10 Hz Frequency (100ms ON 100ms OFF) | LED Module Status Self-Test                                                                 |
|             | Blink with 2.5 Hz (400 ms ON and 400 ms OFF)    | LED Module Standby Activated                                                                |
|             | Steady ON                                       | LED module Operational                                                                      |
| LED A RED   | OFF                                             | Not used                                                                                    |
| LED B GREEN | 1Hz frequency (500ms ON and 500ms OFF)          | No error                                                                                    |
|             | OFF                                             | Device error or in boot mode                                                                |
| LED B RED   | OFF                                             | No error                                                                                    |
|             | Blinking (1Hz, 500ms ON 500ms OFF)              | Warning faults                                                                              |
|             | Fast blinking (2Hz, 250ms ON and 250ms OFF)     | Critical faults                                                                             |
| LED C GREEN | OFF                                             | Not used                                                                                    |
| LED C RED   | OFF                                             | LED Network Status Indicator Steady OFF or LED Network Status Indicator Connection time out |
|             | Blink 400ms                                     | LED Network Status Indicator No Connection                                                  |
|             | ON                                              | LED Network Status Indicator connected                                                      |

## EtherCAT LED Indication details

| LED         | State                                       | Description                  |
|-------------|---------------------------------------------|------------------------------|
| LED A GREEN | OFF                                         | Init                         |
|             | Blinking                                    | Pre-Operational              |
|             | Single Flash                                | Safe-Operational             |
|             | ON                                          | Operational                  |
| LED A RED   | OFF                                         | Not used                     |
| LED B GREEN | 1Hz frequency (500ms ON and 500ms OFF)      | No error                     |
|             | OFF                                         | Device error or in boot mode |
| LED B RED   | OFF                                         | No error                     |
|             | Blinking (1Hz, 500ms ON 500ms OFF)          | Warning faults               |
|             | Fast blinking (2Hz, 250ms ON and 250ms OFF) | Critical faults              |
| LED C GREEN | OFF                                         | Not used                     |
| LED C RED   | OFF                                         | No Error                     |
|             | Blinking                                    | Invalid Configuration        |
|             | Single Flash                                | Unsolicited State Change     |
|             | Double Flash                                | Application Watchdog Timeout |
|             | Flickering                                  | Booting Error                |
|             | ON                                          | PDI Watchdog Timeout         |

### 5.3 DIP Switch: Baud rate & node ID setting (Location 2)

The baud rate and Node ID can be set by SDO or 8 pin DIP switch. The default baud rate is 125 kbps. Possible baud rate supported are 125 kbps, 250 kbps, 500 kbps and 1000 kbps. The upper two MSB pin B7 & B6 are used for baud rate setting as per the table, while remaining 6 pins B5 to B0 are used for Node ID from 1 to 63.

#### DIP switch

Setting of node ID & baud rate:

Switch up = 0

Switch down = 1

Example for setting baud rate & node ID

- To set the Node ID use switch B0 (LSB) to B5 that is 6-bit node address; to set Node ID to 3 make switch B0 & B1 to 1 by pulling down and remaining switched B2 to B5 make 0 by pulling up.
- To set the baud rate use switch B6 & B7 (MSB) that is 2-bit baud rate setting; to set baud rate to 500 kbps make only switch B7 to 1 by pulling down and B6 to 0 by pulling up.

| DIP8/B7 | DIP7/B6 | BAUD RATE |
|---------|---------|-----------|
| 0       | 0       | 125 kbps  |
| 0       | 1       | 250 kbps  |
| 1       | 0       | 500 kbps  |
| 1       | 1       | 1000 kbps |

| B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 | HEX | Baud rate B7/B6 | Node ID B5...B0                          |
|----|----|----|----|----|----|----|----|-----|-----------------|------------------------------------------|
| 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 00  | 125 kbps        | Default (Node ID 1)<br>Node ID = 1....63 |
| 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 05  |                 |                                          |
| 0  | 0  | 1  | 1  | 1  | 1  | 1  | 1  | 3F  |                 |                                          |
| 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 40  | 250 kbps        | Default (Node ID 1)<br>Node ID = 1....63 |
| 0  | 1  | 0  | 0  | 0  | 0  | 0  | 1  | 41  |                 |                                          |
| 0  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 7F  |                 |                                          |
| 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 80  | 500 kbps        | Default (Node ID 1)<br>Node ID = 1....63 |
| 1  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 81  |                 |                                          |
| 1  | 0  | 1  | 1  | 1  | 1  | 1  | 1  | BF  |                 |                                          |
| 1  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | C0  | 1000 kbps       | Default (Node ID 1)<br>Node ID = 1....63 |
| 1  | 1  | 0  | 0  | 0  | 0  | 0  | 1  | C1  |                 |                                          |
| 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0  | FE  |                 |                                          |

### 5.4 Micro USB serial communication (Location 2)

The micro USB 2.0 port is not intended for a permanent installation when the valve body and the computer both have connections to protective earth. The USB ground is not isolated, so either the computer should run on a battery (laptop) or the valve should not be connected to ground. If that cannot be achieved, please use a USB isolator between the computer and the valve. Also consider that the IP67 is compromised when the PG plug is removed and the USB connection used.

**Recommended cable type:** Standard "A" Male to Micro-B Male.

**Note:** It is recommended to use a USB isolator to eliminate any ground loops through the serial USB port.

## 5.5 Electromagnetic compatibility (EMC)

|                                                                               |                                                                                                                                    |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Electromagnetic compatibility (EMC):                                          | IEC61326-1 2021                                                                                                                    |
| Conducted Emissions CISPR 11                                                  | 2024/EN55011- Class A, 150kHz - 30MHz.                                                                                             |
| Radiated Emissions CISPR 11                                                   | 2024/EN55011- Class A, 30MHz – 6GHz.                                                                                               |
| RF Continuous Conducted Disturbances (Conducted Susceptibility) IEC 61000-4-6 | 2023, Class A 150KHz to 80MHz <ul style="list-style-type: none"> <li>DC Power Port: 10Vrms</li> <li>Signal Port: 10Vrms</li> </ul> |
| RF Electromagnetic Fields (Radiated Susceptibility) IEC 61000-4-3             | 2020, Class A<br>80MHz-1000MHz: (10V/m)<br>1.4GHz-6GHz: (3V/m)                                                                     |
| Surges IEC 61000-4-5                                                          | 2014+A1: 2017, Class B                                                                                                             |
| DC Power Port                                                                 | ±0.5kV Line to Line, ±1kV Line to Ground                                                                                           |
| Signal Port                                                                   | ±1kV Line to Ground                                                                                                                |
| Electrical Fast Transients IEC 61000-4-4                                      | 2012, Class B <ul style="list-style-type: none"> <li>DC Power Port: ±2kV</li> <li>Signal Port: ±1kV</li> </ul>                     |
| Electrostatic Discharges (ESD) IEC 61000-4-2                                  | 2008, Class B <ul style="list-style-type: none"> <li>Air discharges: ±8kV</li> <li>Contact discharges: ±4kV</li> </ul>             |
| Power Frequency Magnetic Fields IEC 61000-4-8                                 | 2009, Class A <ul style="list-style-type: none"> <li>(30A/m), Continuous field</li> </ul>                                          |
| Voltage Withstand Test                                                        | 750V DC for 60sec                                                                                                                  |

## 5.6 Operating Conditions

|                                                                                                                                 |                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Monitor Output Signal:</b><br>Voltage mode<br>Voltage Mode Output Impedance<br>Current Mode<br>Current Mode Output Impedance | ±10V DC<br>10 kΩ<br>4-20mA<br>0-250Ω (typ 100Ω)                                                                                  |
| Power stage PWM frequency                                                                                                       | 21 kHz nominal                                                                                                                   |
| Reproducibility, valve-to-valve (at factory settings):<br>Flow gain at 100% command signal                                      | ≤5%                                                                                                                              |
| <b>Protection:</b><br>Electrical                                                                                                | Reverse polarity protected between pin A-B of the 7 pin plug connector and between pin 1-2 and 9-10 of the 12 pin plug connector |
| Ambient air temperature range for full performance                                                                              | –40°C to +85°C (–40°F to +185°F)                                                                                                 |
| Oil temperature range for full performance                                                                                      | 0°C to 70°C (32°F to 158°F)                                                                                                      |
| Minimum temperature at which valves will work at reduced performance                                                            | –40°C (–40°F)                                                                                                                    |
| Storage temperature range                                                                                                       | –25°C to +85°C (–13°F to +185°F)                                                                                                 |
| <b>Vibration:</b><br>Random Vibration (5 Hz to 2000 Hz, 10 hours per axis)<br>Mechanical shock                                  | 13.25 Grms<br>50 G, 6 positive and 6 negative pulses per axis                                                                    |
| Power supply                                                                                                                    | 24V DC (18V to 36V including 10% peak-to-peak max ripple) max current 3.2A                                                       |
| <b>Command Signal:</b><br>Voltage mode<br>Input impedance<br>Current mode                                                       | –10V to +10V DC<br>Field 5 = 1: 47kΩ, Field 5 = 2,3,4,5: 100Ω<br>Field 5 = 2,3,4,5                                               |
| Valve Enable Signal for model code field 7<br>Enable Disable<br>Input impedance                                                 | Disable <5V Enable Signal >12V (max 36V)<br>10 kΩ                                                                                |
| Amplifier Temperature Sensing                                                                                                   | 1°C (1.8°F) resolution, –25°C (–13°F) undertemp detect, 125°C (257°F) overtemp detect                                            |
| Power Supply Detect                                                                                                             | 18-36V DC, 0.01 V resolution ± 1%, 18V under voltage detect, 36V overvoltage                                                     |

|                                                                                                                    |                                                                    |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>KBHD15-07 Valves (all valves)</b><br>Relative duty factor<br>Hysteresis                                         | Continuous rating (ED = 100%)<br><0.4%                             |
| <b>Mass:</b><br>Valves with pressure reducer<br>Valves without pressure reducer                                    | <b>kg (lb) approx.</b><br>11 kg(24lb) approx<br>10 kg(23lb) approx |
| <b>Step response</b><br>Step, % Flow<br>0% to 100%, 100% to 0%<br>10% to 90%, 90% to 10%<br>25% to 75%, 75% to 25% | ms<br><60<br><50<br><45                                            |
| <b>Parts Information:</b><br>Mating Electrical 7-pin Connector<br>Mating Electrical 12-pin Connector               | 934939<br>12007132                                                 |

| MTTFd, refer EN 13849 data sheet DOC00092051                                                                                                                 | Condition                                   | MTTFd(Years) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------|
| KBHD15-07-*****-B/E/C/D-**-**07-NS/CO-N-01-H-001-**-*****<br>(Valves with 7 Pin connector and Analog or CANopen communication option, without reducer)       | Power removed (Switch off)<br>Power applied | 75<br>49.9   |
| KBHD15-07-*****-B/E/C/D-**-**07-PN/EI/EC-N-01-H-001-**-*****<br>(Valves with 7 Pin connector and Industrial ethernet communication option, without reducer)  | Power removed (Switch off)<br>Power applied | 75<br>40.3   |
| KBHD15-07-*****-B/E/C/D-**-**12-NS/CO-N-01-H-001-**-*****<br>(Valves with 12 Pin connector and Analog or CANopen communication option, without reducer)      | Power removed (Switch off)<br>Power applied | 75<br>48.6   |
| KBHD15-07-*****-B/E/C/D-**-**12-PN/EI/EC-N-01-H-001-**-*****<br>(Valves with 12 Pin connector and Industrial ethernet communication option, without reducer) | Power removed (Switch off)<br>Power applied | 75<br>39.4   |
| KBHD15-07-*****-A/F/G/H-**-**07-NS/CO-N-01-H-001-**-*****<br>(Valves with 7 Pin connector and Analog or CANopen communication option, with reducer)          | Power removed (Switch off)<br>Power applied | 50<br>37.4   |
| KBHD15-07-*****-A/F/G/H-**-**07-PN/EI/EC-N-01-H-001-**-*****<br>(Valves with 7 Pin connector and Industrial ethernet communication option, with reducer)     | Power removed (Switch off)<br>Power applied | 50<br>31.8   |
| KBHD15-07-*****-A/F/G/H-**-**12-NS/CO-N-01-H-001-**-*****<br>(Valves with 12 Pin connector and Analog or CANopen communication option, with reducer)         | Power removed (Switch off)<br>Power applied | 50<br>36.7   |
| KBHD15-07-*****-A/F/G/H-**-**12-PN/EI/EC-N-01-H-001-**-*****<br>(Valves with 12 Pin connector and Industrial ethernet communication option, with reducer)    | Power removed (Switch off)<br>Power applied | 50<br>31.2   |

## 6. Software Information

- Analog commanded spool control.
- Analog command source configuration options.
- Monitor output signal configuration options.
- Enable input signal enable/disable option.
- Valve spool position control (VPOC/VSC)
- CANopen DSP306 compliant electronic data sheet (EDS).
- Diagnostic configuration options.
- Ethernet Fieldbus option: Profinet, Ethernet/IP, EtherCAT
- TCP/IP communication for Danfoss Pro-FX™ 2.0 configuration tool - closed loop control modes are available via industrial Ethernet

All levels and models are compatible with the Danfoss Pro-FX™. For the latest revision, please visit [www.danfoss.com](http://www.danfoss.com)

**Download Pro-FX™, Technical Information and Support Materials from Danfoss's Website: <http://www.danfoss.com>**

Install the Danfoss Pro-FX™ Configure PC application tool. Installation is supported on a wide range of Windows based operating systems including Windows 10/11 32 bit and 64 bit.

The Pro-FX™ Configure installation provides several options for PC USB peripheral CANbus adapters supported by the software. During installation the user can choose to install drivers for an available CANbus adapter.

The adapters supported by Pro-FX™ Configure are:

- PCAN-USB\* PEAK-System Technik GmbH (<http://www.peak-system.com>)
- ValueCAN Intrepid Control Systems, Inc. (<http://www.intrepidcs.com>)
- Leaf-Light Kvaser AB (<http://www.kvaser.com>)

\* The PCAN-USB adapter is recommended for compatibility with Danfoss Pro-FX™. Control development environment used with KBHD15-07 and other Danfoss Pro-FX™ products, it is also included in the TEQ-470-A-10 test box.

The adaptors supported by Pro-FX™ Configure are:

CG150  
TCP/IP  
USB Serial

### Profinet Version:

- Protocol IO-Device
- Cyclic 5ms data exchange
- Conformance Class 2
- PROFINET IO specification: V 2.4
- 100/10 Mbit RT compatible
- Acyclic data exchange via Read Write Data Records
- Two ports both can be used as access, build in Switch
- GDML file for Level1 and Level2
- TCP ports:
  - 80 (webserver)
  - 300 (Pro-FX™ tool)
- Pro-FX™ tool access via TCP/IP and Serial

### CANopen Version:

- CANopen Slave Device
- PDO communication with dynamic mapping support
- Baud rate supported: 125kbps, 250kbps, 500kbps and 1000kbps
- SDO communication
- EDS file available
- Pro-FX™ tool access via CANopen and serial

### EtherCAT Version:

- EtherCAT Slave Device
- PDO communication 100µs ring time 5ms refresh rate
- 100/10Mbit data rate
- SDO communication (CoE)
- PDO dynamic mapping support
- ESI file available
- FW download via FoE supported
- Pro-FX™ tool access via serial

### Ethernet/IP Version:

- Ethernet/IP Adapter Device
- Assembly package for Process Data 5ms refresh rate
- 100/10Mbit data rate
- two ports supported with daisy chaining, DLR support
- Class 3 explicit message support
- EDS file available
- TCP ports:
  - 80 (webserver)
  - 300 (Pro-FX™ tool)
- Pro-FX™ tool access via TCP/IP and Serial

## 7. Electrical Information

### 7.1 Block Diagram Voltage Input (Field 5=1)

Wiring connections must be made via the 7-pin plug or 12-pin plug mounted on the amplifier. See page 22 & 23 of this user guide.

Recommended cable sizes are:

#### Power cables:

For 24V supply

0.75 mm<sup>2</sup> (18 AWG) up to 20m (65 ft)

1.00 mm<sup>2</sup> (16 AWG) up to 40m (130 ft)

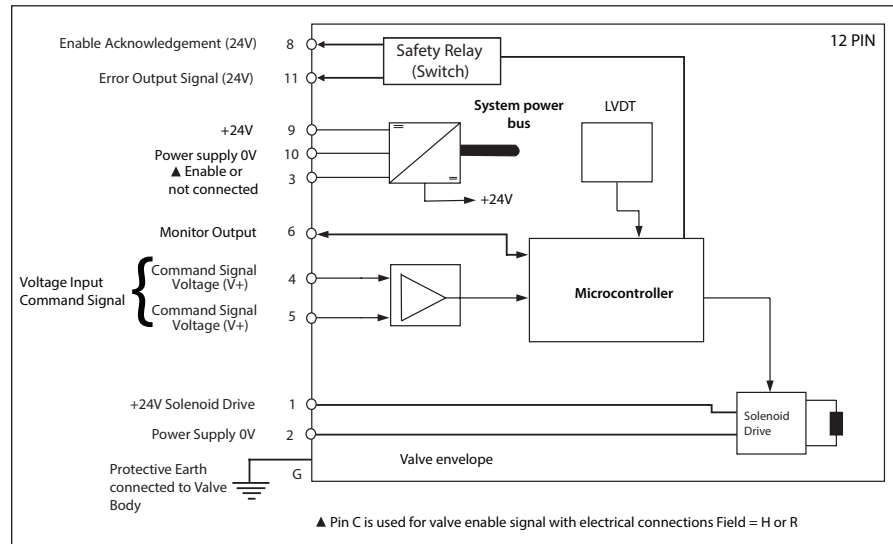
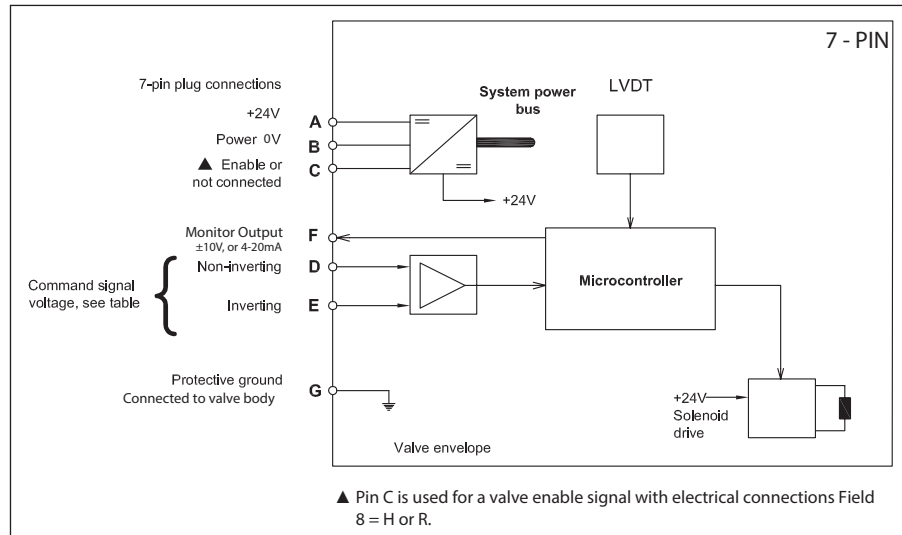
#### Signal cables:

0.50 mm<sup>2</sup> (20 AWG)

#### Screen (shield):

A suitable cable would have 7 cores or 10/12 cores, a separate screen for the signal wires and an overall screen. Cable outside diameter 8.0 - 10.5 mm (0.31 - 0.41 inches) See connection diagram on page 22 & 23.

#### Voltage Input Mode



#### Command Signals and Outputs, Field 5 = 1

| 7-pin plug / 12-pin plug                        |                                         | Flow Direction |
|-------------------------------------------------|-----------------------------------------|----------------|
| Pin D (7-pin plug)/ Pin 4 (12-pin plug)         | Pin E (7-pin plug)/ Pin 5 (12-pin plug) |                |
| Positive<br>0V<br>$U_D - U_E = \text{Positive}$ | 0V<br>Negative                          | P to A         |
| Negative<br>0V<br>$U_D - U_E = \text{Negative}$ | 0V<br>Positive                          | P to B         |

## 7.2 Block Diagram Current Input (Field 5= 2,3,4,5)

Wiring connections must be made via the 7-pin plug or 12-pin plug mounted on the amplifier. See page 24 & 25 of this user guide.

Recommended cable sizes are:

### Power cables:

For 24V supply

0.75 mm<sup>2</sup> (18 AWG) up to 20m (65 ft)

1.00 mm<sup>2</sup> (16 AWG) up to 40m (130 ft)

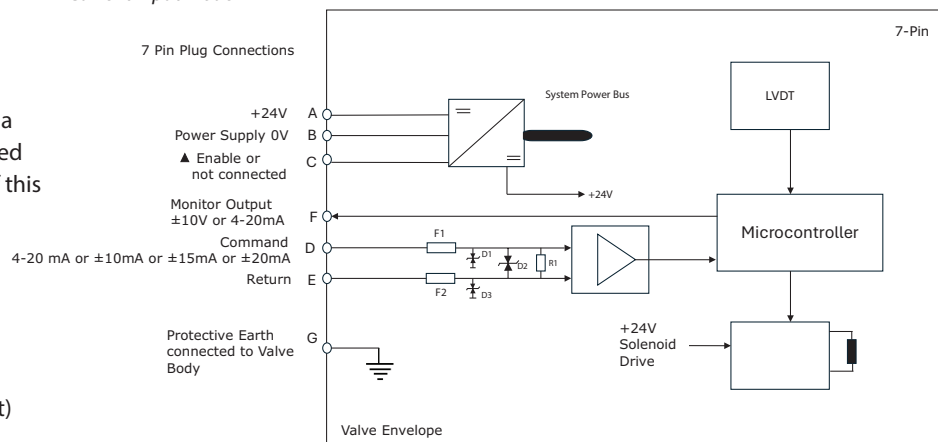
### Signal cables:

0.50 mm<sup>2</sup> (20 AWG)

### Screen (shield):

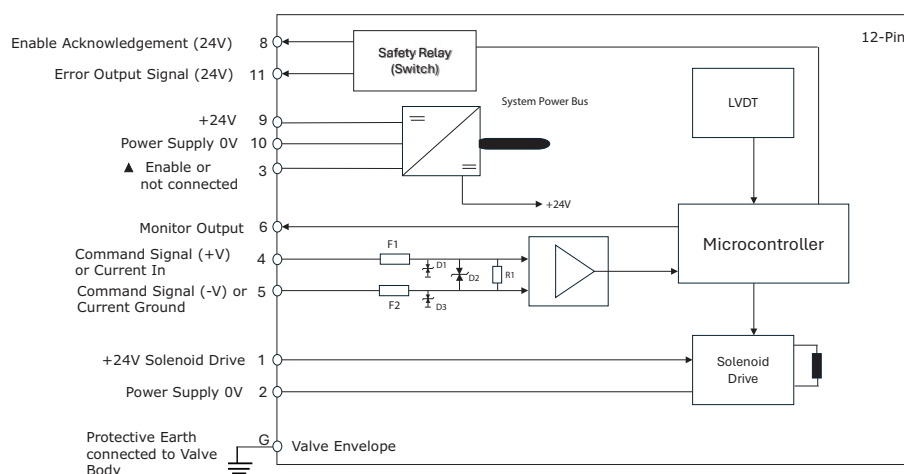
A suitable cable would have 7 cores or 10/12 cores, a separate screen for the signal wires and an overall screen. Cable outside diameter 8.0 - 10.5 mm (0.31 - 0.41 inches) See connection diagram on page 24 & 25.

### Current Input Mode



▲ Pin C is used for a valve Enable signal with Electrical connections Field = H or R

R1 Shunt Resistor 100R  
F1,F2 Resettable Fuse  
Dx TVS Diode rated for 15V



▲ Pin C is used for a valve Enable signal with Electrical connections Field = H or R

R1 Shunt Resistor 100R  
F1,F2 Resettable Fuse  
Dx TVS Diode rated for 15V

### Command Signals and Outputs, Field 5=2

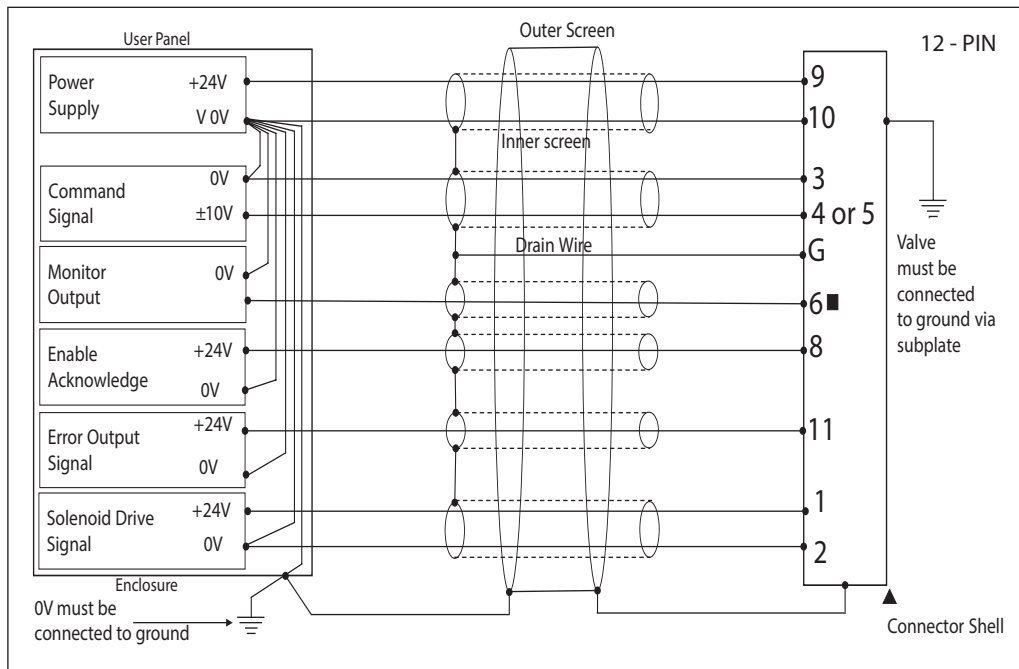
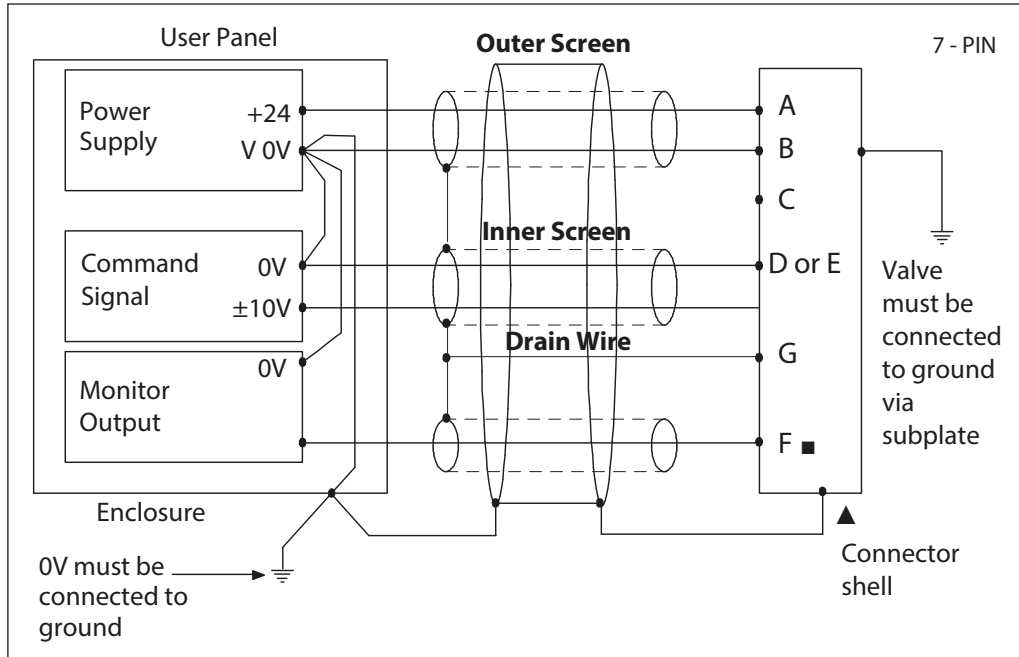
| 7-pin plug / 12-pin plug                |                                         |                                         | Flow Direction |
|-----------------------------------------|-----------------------------------------|-----------------------------------------|----------------|
| Pin D (7-pin plug)/ Pin 4 (12-pin plug) | Pin E (7-pin plug)/ Pin 5 (12-pin plug) | Pin B (7-pin plug)/ Pin 2 (12-pin plug) |                |
| More than 12 mA                         | Current return                          | Power ground                            | P to A         |
| Less than 12 mA                         | Current return                          | Power ground                            | P to B         |

### Command Signals and Outputs, Field 5=3,4,5

| 7-pin plug / 12-pin plug                |                                         |                                         | Flow Direction |
|-----------------------------------------|-----------------------------------------|-----------------------------------------|----------------|
| Pin D (7-pin plug)/ Pin 4 (12-pin plug) | Pin E (7-pin plug)/ Pin 5 (12-pin plug) | Pin B (7-pin plug)/ Pin 2 (12-pin plug) |                |
| More than 0 mA                          | Current return                          | Power ground                            | P to A         |
| Less than 0 mA                          | Current return                          | Power ground                            | P to B         |

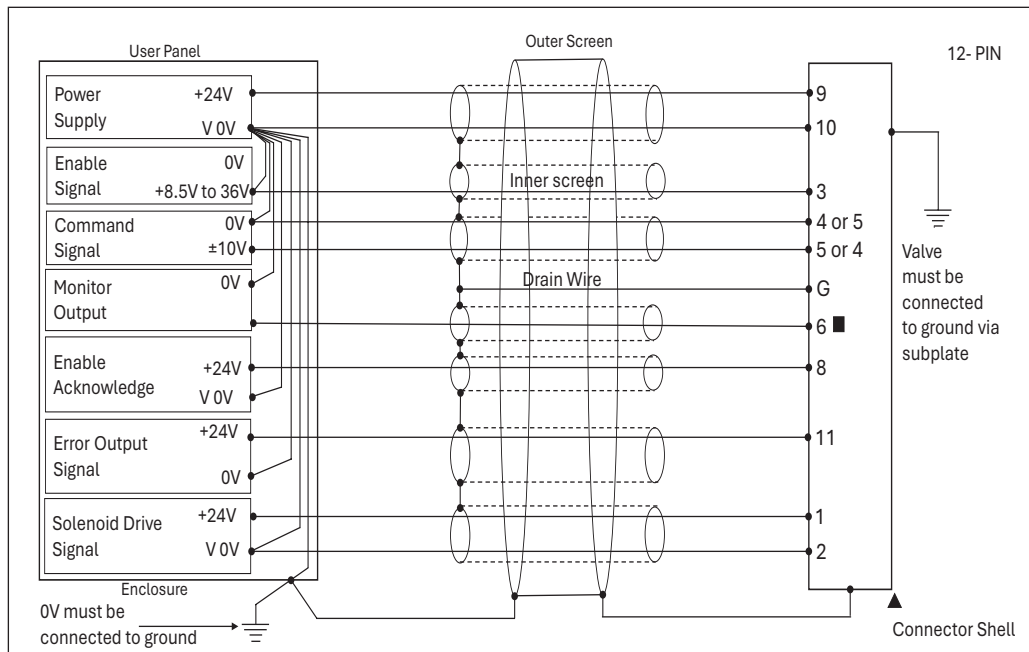
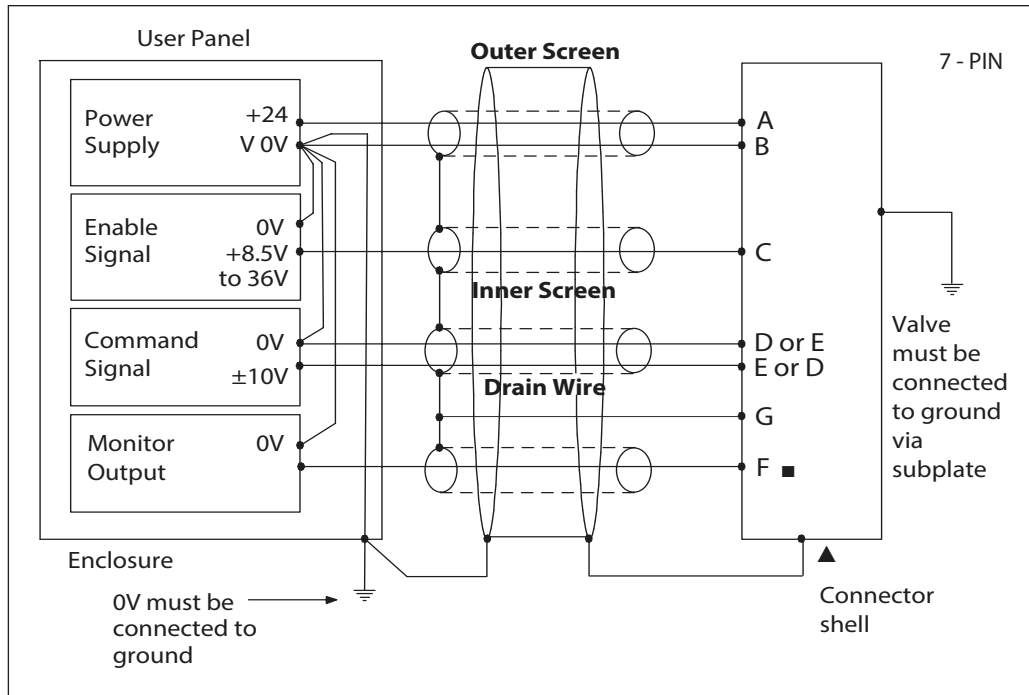
### 7.3 Wiring Connections Voltage Input (Field 5 = 1)

- Spool position monitor voltage (pin F/ pin 6) will be referenced to the valve local ground (pin B/ pin 10).



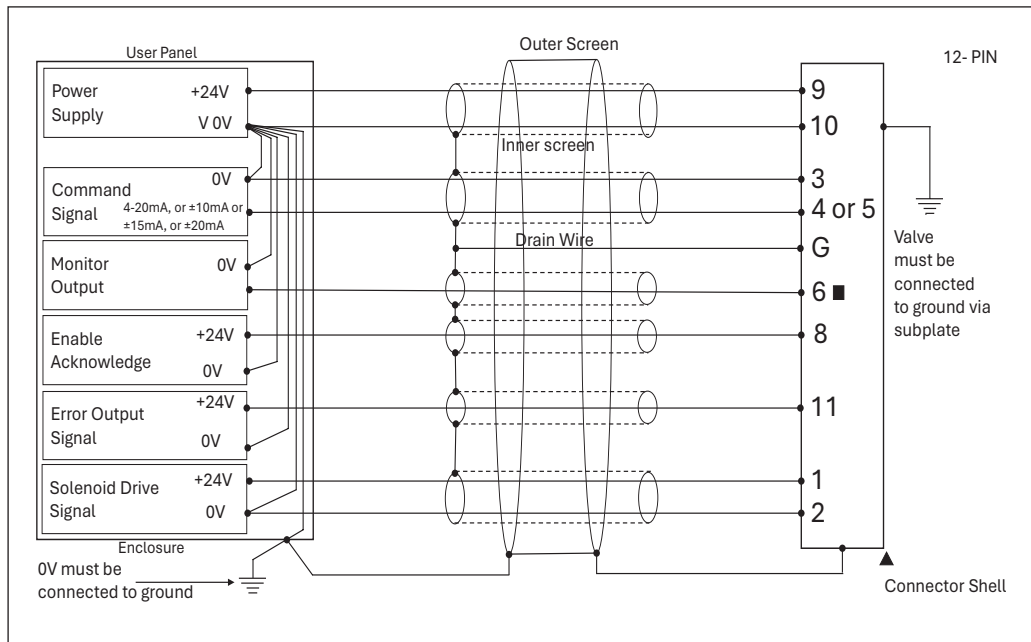
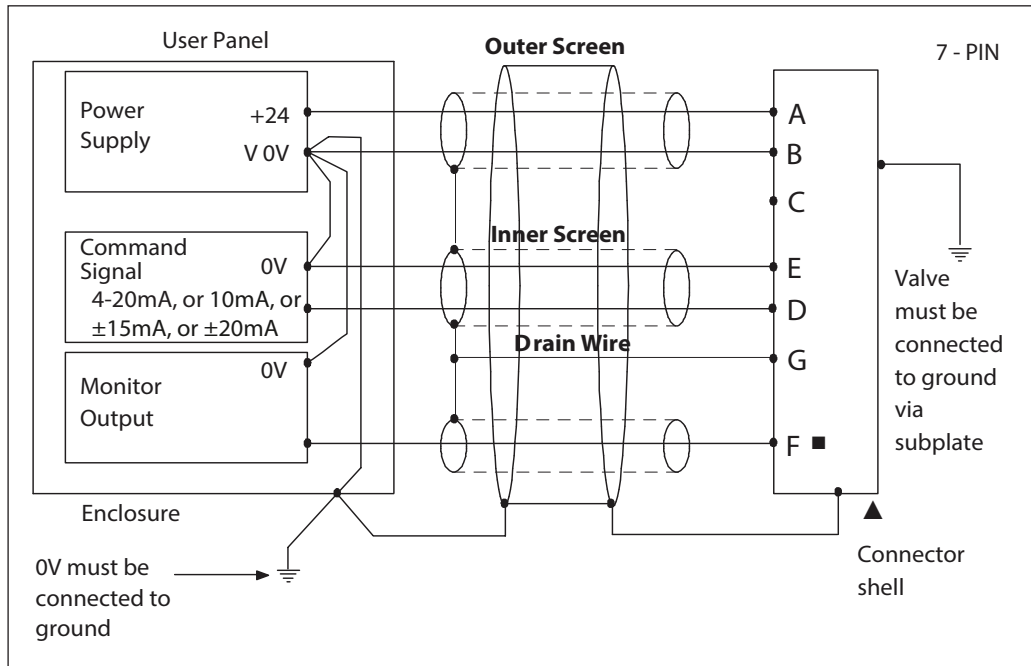
## 7.4 Wiring Connections for Voltage Mode (Field 5 = 1) Valves with Enable Feature

ΔNote: In applications where the valve must conform to European RFI/EMC regulations, the outer screen (shield) must be connected to the outer shell of the 7 pin connector, and the valve body must be fastened to the earth ground. Proper earth grounding practices must be observed in this case, as any differences in command source and valve ground potentials will result in a screen (shield) ground loop.



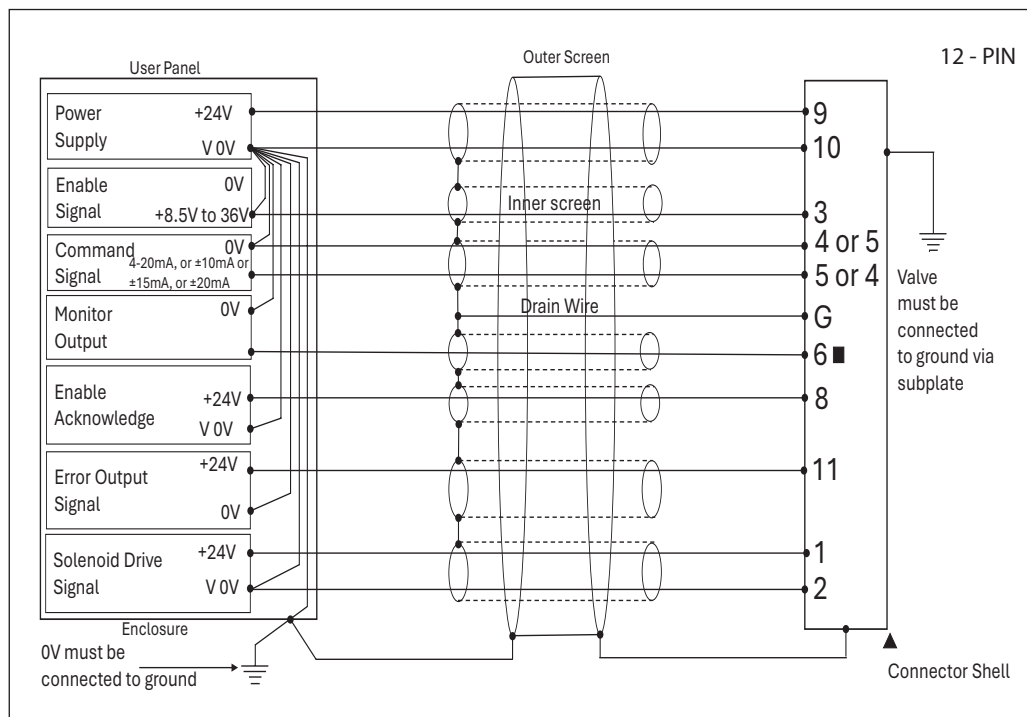
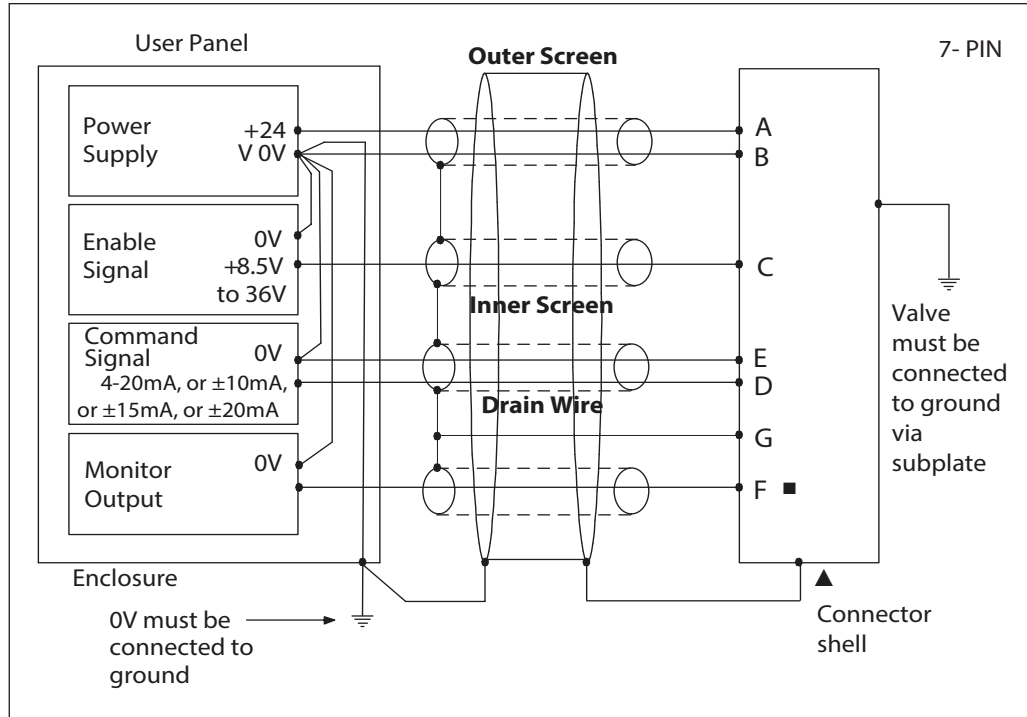
### 7.5 Wiring Connections Current Input (Field 5 = 2,3,4,5)

- Spool position monitor voltage (pin F) will be referenced to the KB valve local ground (pin B).



## 7.6 Wiring Connections for Current Input (Field Field 5 = 2,3,4,5) Valves with Enable Feature

Δ Note: In applications where the valve must conform to European RFI/EMC regulations, the outer screen (shield) must be connected to the outer shell of the 7 pin/ 12 pin connector, and the valve body must be fastened to the earth ground. Proper earth grounding practices must be observed in this case, as any differences in command source and valve ground potentials will result in a screen (shield) ground loop.

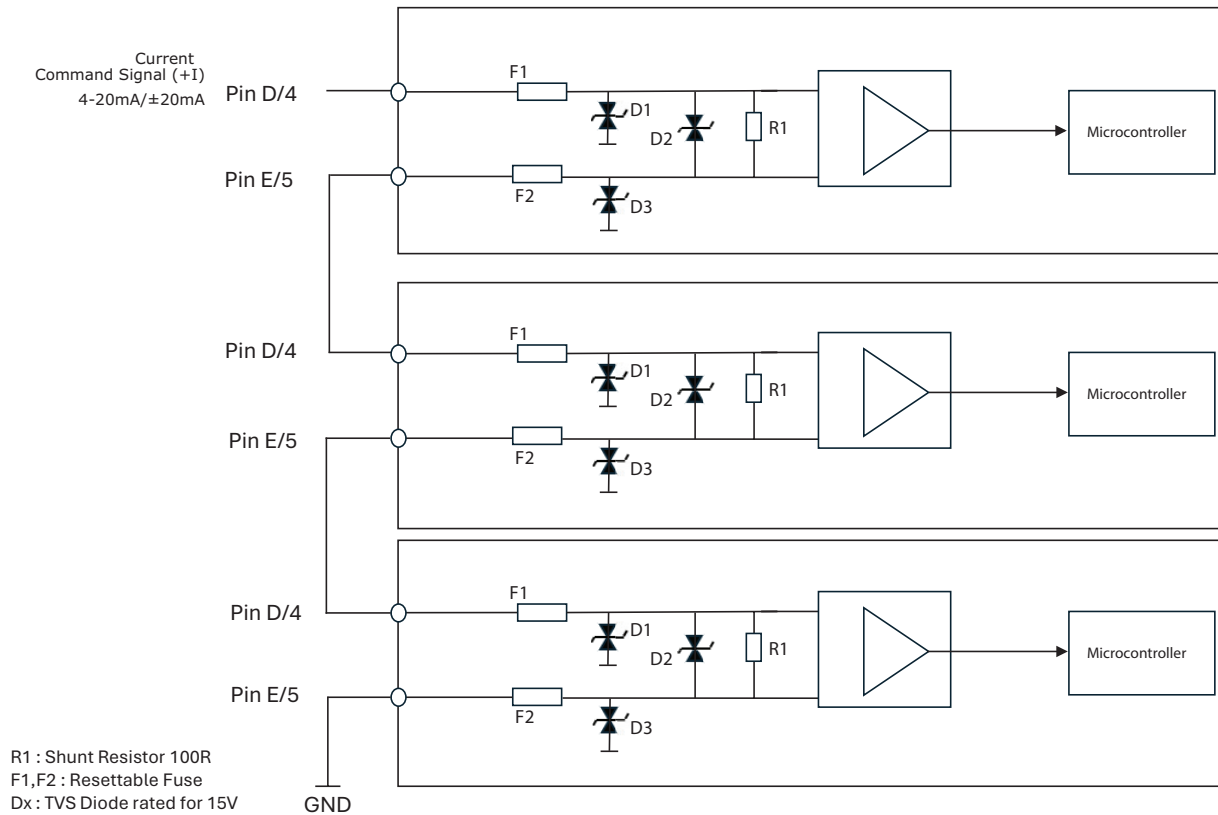


### 7.7 Wiring Connection of Valve in Current Mode Command with current as input

#### Daisy Chain Configuration

In applications where valves are connected in a daisy chain configuration, the following wiring method should be used:

- Pin D/4 of the valve serves as the input.
- Pin E/5 acts as the return.
- The E/5 pin of the first valve should be connected to the D/4 pin of the next valve in the series.
- This pattern continues for all valves in the daisy chain.



**Note:** Detailed wiring connections are outlined in Section 7.5 and 7.6

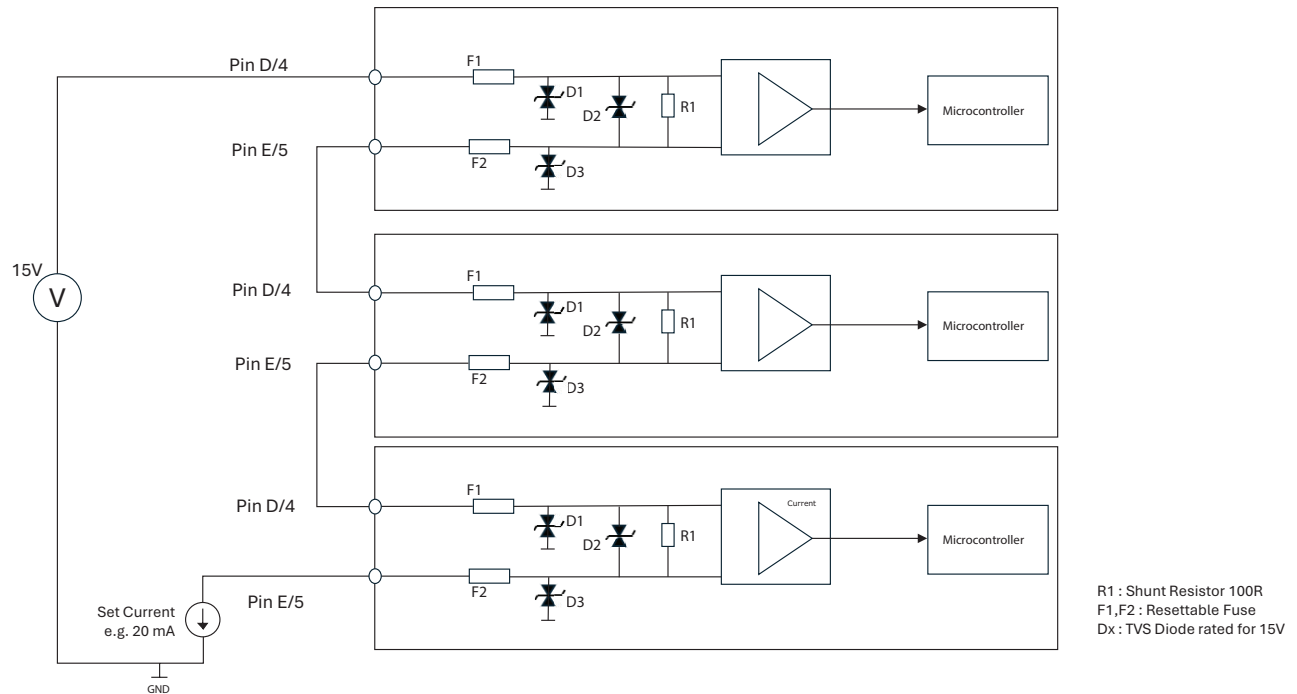
The daisy chain configuration above is applicable to both 7 Pin and 12 Pin Connector

## 7.8 Wiring Connection of Valve in Current Mode Command with voltage as input

### Daisy Chain Configuration

In applications where valves are connected in a daisy chain configuration, the following wiring method should be used:

- Apply 15V (maximum) input to the valve on Pin D/4.
- Pin E/5 acts as the return.
- This wiring pattern continues for all valves in the daisy chain.
- At the termination point (i.e., Pin E/5), the adjustment of current drawn determines the current flowing through the valves in series.



**Note:** Detailed wiring connections are outlined in Section 7.5 and 7.6

The daisy chain configuration above is applicable to both 7 Pin and 12 Pin Connector

## 7.9 Warning



Electromagnetic Compatibility (EMC) It is necessary to ensure that the valve is wired up as above. For effective protection of the user electrical cabinet, the valve subplate or manifold and the cable screens should be connected to efficient ground points. The metal 7 pin connector part no. 934939 or 12 pin connector part no. 12007132 should be used for the integral amplifier.

In all cases both valve and cable should be kept as far away as possible from any sources of electromagnetic radiation such as cables carrying heavy current, relays and certain kinds of portable radio transmitters, etc.

Difficult environments could mean that extra screening may be necessary to avoid the interference. It is important to connect the 0V lines as shown above. The multi-core cable should have at least two screens to separate the demand signal and monitor output from the power lines.

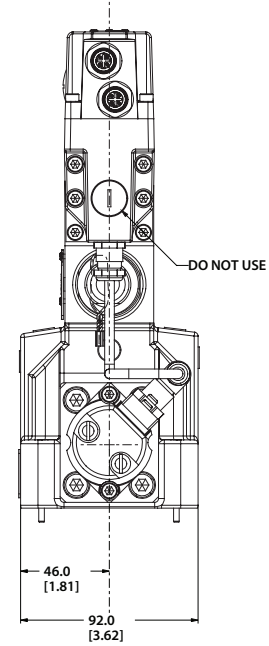
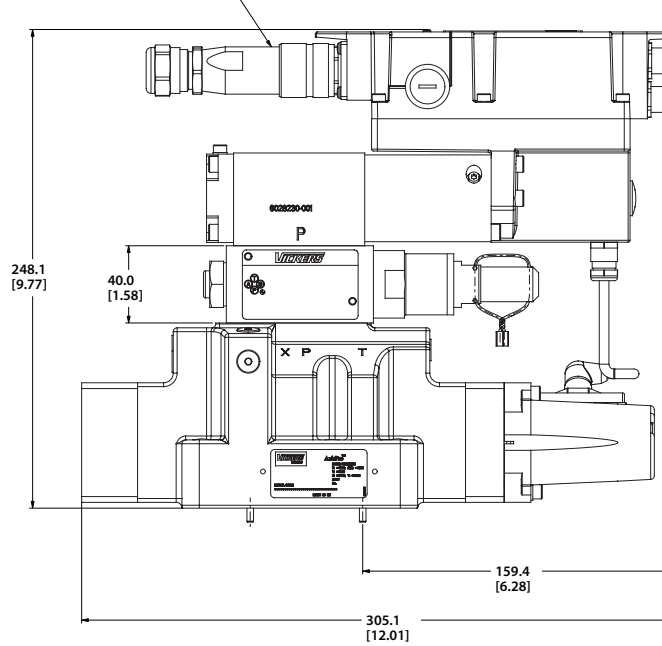
The enable line to pin C should be outside the screen which contains the demand signal cables.

To ensure EMI protection use only metal shielded mating connectors.

## 8. Installation Dimensions

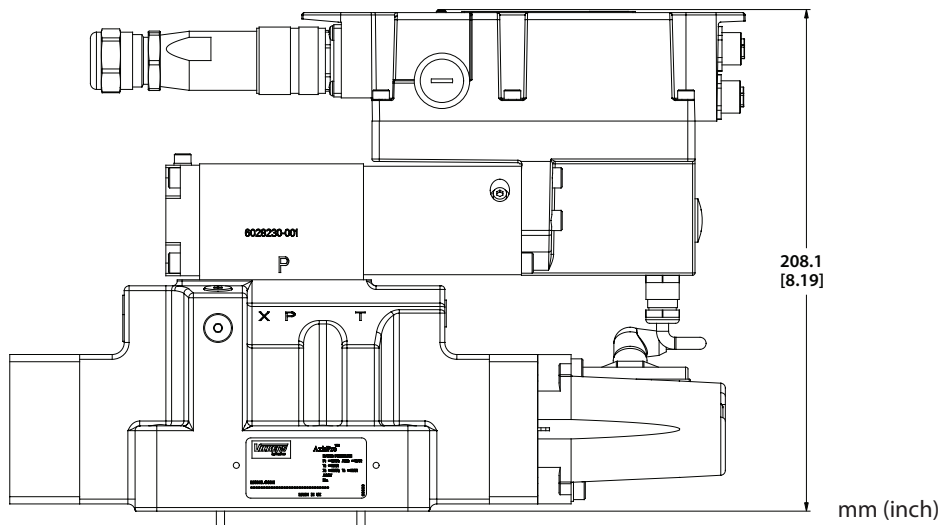
KBHD15-07 with Pressure Reducer

7 PIN/12 PIN CONNECTOR PLUG



mm (inch)

KBHD15-05 without Pressure Reducer



mm (inch)

#### A. Mounting bolt

- Metric : M10 X 50 - Quantity : 4  
M6 X 50 - Quantity : 2
  - Inch : 3/8"-16 UNC - Quantity : 4  
2/4"-20 UNC - Quantity : 2
- Recommended blot : Class 12.9 (ISO 898)  
Recommended torque : M10 - 50 to 60 Nm  
M6 - 11 to 15 Nm

#### B. The connector plug

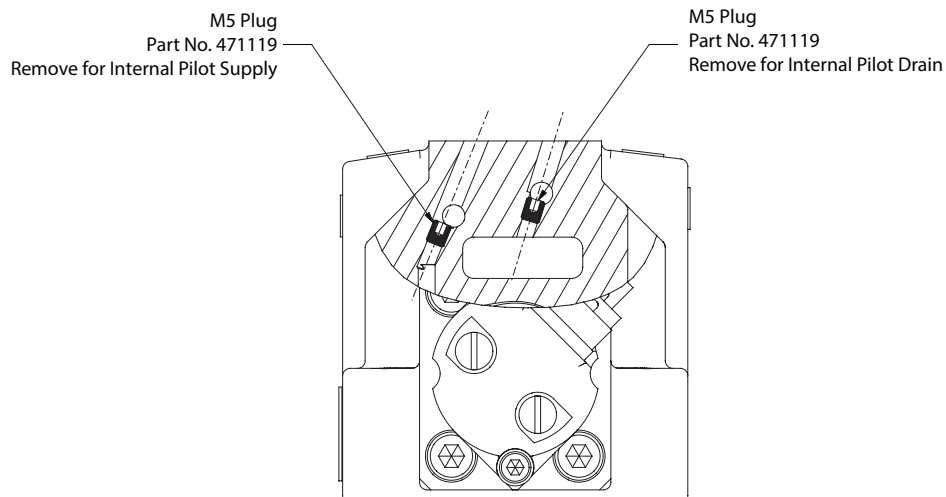
- 7 Pin connector : 934939
- 12 Pin connector : 12007132

Connector plug retaining nut must be tightened with a torque of 2.0 to 2.5 Nm to effect a proper seal.

This is necessary to ensure that the EMC rating and IP67 rating are achieved.

#### Pilot Supply and Drain Plugs

Viewed from port A end of main-stage



Internal plugs shown, for external pilot supply (via port X) and external drain (via port Y).

For internal pilot supply (from port P), plug is not fitted. Port X should be blocked at mounting interface or otherwise plugged at subplate of manifold block.

Similarly, for internal pilot drain (through port T), plug is not fitted. Port Y should be blocked at mounting interface or otherwise plugged at subplate of manifold block.

## 9. Mounting Surfaces

Mounting Surface Interface to ISO 4401 (Size 07)

This interface conforms to: ISO 4401-07-07-0-05

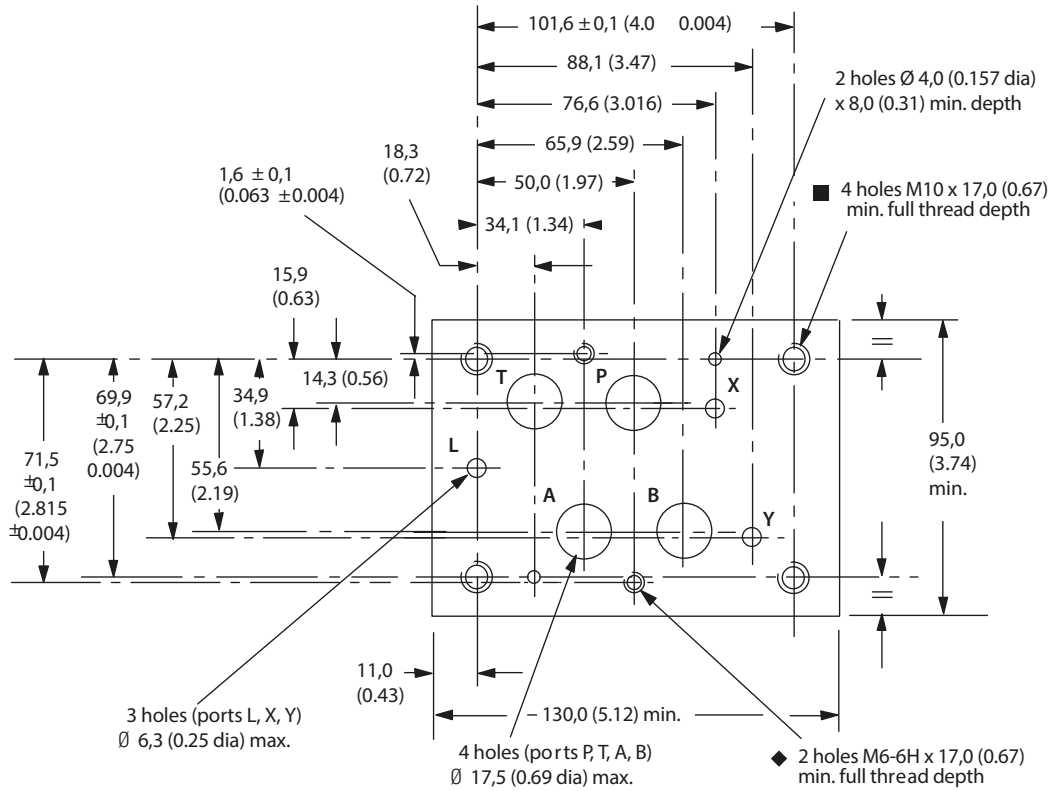
ANSI/B93.7M (and NFPA) size 07

CETOP R35H4.3-07

DIN 24340 Form A16

■ - 3/8 -16 UNC optional.

◆ - 1/4 -20 UNC optional.



## 10. Application Data

### Hydraulic Fluids

Proper selection of hydraulic fluid and consistent maintenance of its condition are essential for ensuring the long and reliable operation of hydraulic components and systems.

The hydraulic fluid must possess a well-balanced formulation of high-quality base oils and additives to provide robust protection against component wear, as well as deliver essential properties like oxidative and thermal stability and rapid air release.

Materials and seals used in these valves are Fluorocarbon (FKM) and are compatible with following hydraulic fluids.

### Mineral/Petroleum oil

- HH, HL, HM, HR, HV, HS & HG (ISO 11158/ISO 6743-4)

### Synthetic PAO (Poly Alfa Olefin)

#### Environment friendly (bio-degradable) hydraulic fluids (ISO 15380/ISO 6743-4)

- HEES (Synthetic ester base))
- HETG (Vegetable oil/triglyceride base)
- HEPR (Polyalphaolefin and related hydrocarbon)

### Fire resistant hydraulic fluids (ISO 12922/ISO 6743-4)

#### Anhydrous/water free

- HFDR (Phosphate ester)
- HFDU (Polyol ester)
- HFDU (Polyether polyol)

### Fluid Cleanliness

The hydraulic fluid must maintain an adequate level of cleanliness at all times to ensure trouble-free performance and prevent premature failure of system components.

The following recommendations are based on ISO 4406:1999 cleanliness standards, which classify particle contamination at 4 µm, 6 µm, and 14 µm sizes. For the products featured in this catalogue, the recommended cleanliness level is 17/15/12.

Danfoss products, as any components, will operate with apparent satisfaction in fluids with higher cleanliness codes than those described. Experience has shown, however, that the life of any hydraulic components is shortened in fluids with higher cleanliness codes than the recommended level.

### Viscosity Recommendations

The extreme operating viscosity range is 500 to 13 cSt (2270 to 70 SUS), but the recommended running range is 54 to 13 cSt (245 to 70 SUS).

### Installation

The proportional valves in this catalog can be mounted in any attitude, but it may be necessary in certain demanding applications, to ensure that the solenoids are kept full of hydraulic fluid.

Good installation practice dictates that the tank port and any drain port are piped so as to keep the valves full of fluid once the system start-up has been completed.

### Service Information

The products from this range are preset at the factory for optimum performance; disassembling critical items would destroy these settings. It is therefore recommended that should any mechanical or electronic repair be necessary they should be returned to the nearest Danfoss repair center.

The products will be refurbished as necessary and retested to specification before return. Field repair is restricted to the replacement of the interface seals.

### Cartridge valves

- DCV directional control valves
- Electric converters
- Electric machines
- Electric motors
- Fluid Conveyance
- Gear motors
- Gear pumps
- Hydraulic integrated circuits (HICs)
- Hydrostatic motors
- Hydrostatic pumps
- Industrial hydraulics
- Orbital motors
- PLUS+1® controllers
- PLUS+1® displays
- PLUS+1® joysticks and pedals
- PLUS+1® operator interfaces
- PLUS+1® sensors
- PLUS+1® software
- PLUS+1® software services, support and training
- Position controls and sensors
- PVG proportional valves
- Steering components and systems
- Telematics

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#### Danfoss

##### Power Solutions (US) Company

2800 East 13th Street  
Ames, IA 50010, USA  
Phone: +1 515 239 6000

#### Danfoss

##### Power Solutions GmbH & Co. OHG

Krokamp 35  
D-24539 Neumünster, Germany  
Phone: +49 4321 871 0

#### Danfoss

##### Power Solutions ApS

Nordborgvej 81  
DK-6430 Nordborg, Denmark  
Phone: +45 7488 2222

#### Danfoss

##### Power Solutions Trading (Shanghai) Co., Ltd.

Building #22, No. 1000 Jin Hai Rd Jin Qiao,  
Pudong New District Shanghai, China 201206  
Phone: +86 21 2080 6201

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